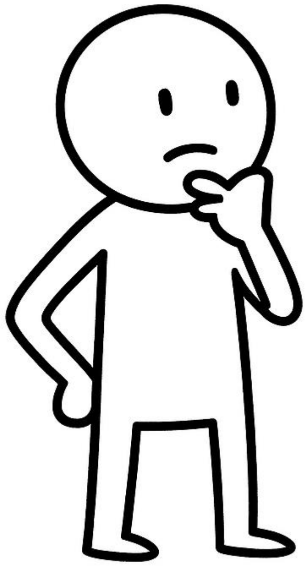
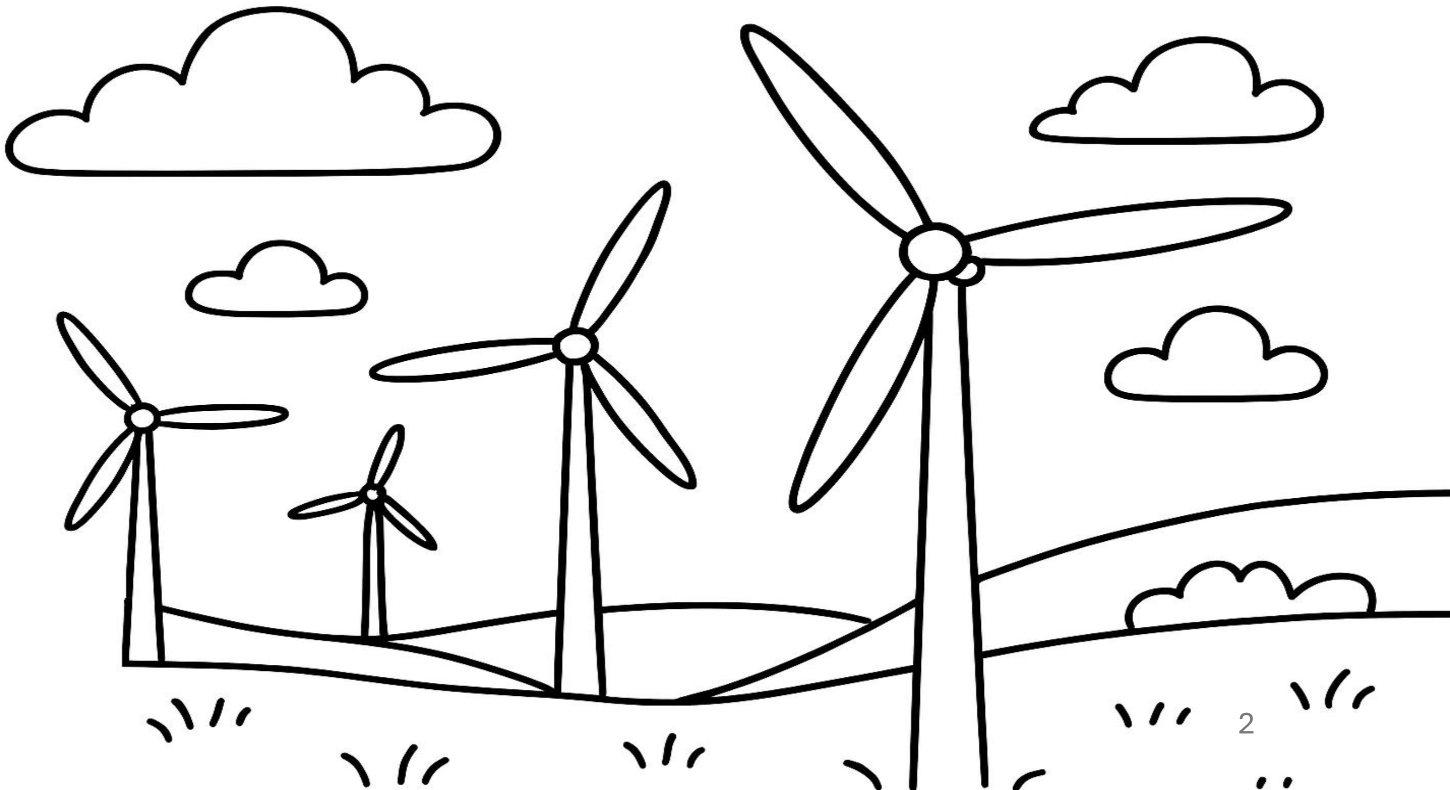
A platform for supporting onshore windfarm siting in Australia

Supervisors: Simone Marinai, Tim Dwyer

Problem Introduction

Where should we build a new windfarm?





Problem Introduction

Factors have different units, scales, distributions, and importance

Technical & Economic

- Wind speed
- Life Cycle Costs
- ROI
- Electricity price and demand

Infrastructure & Accessibility

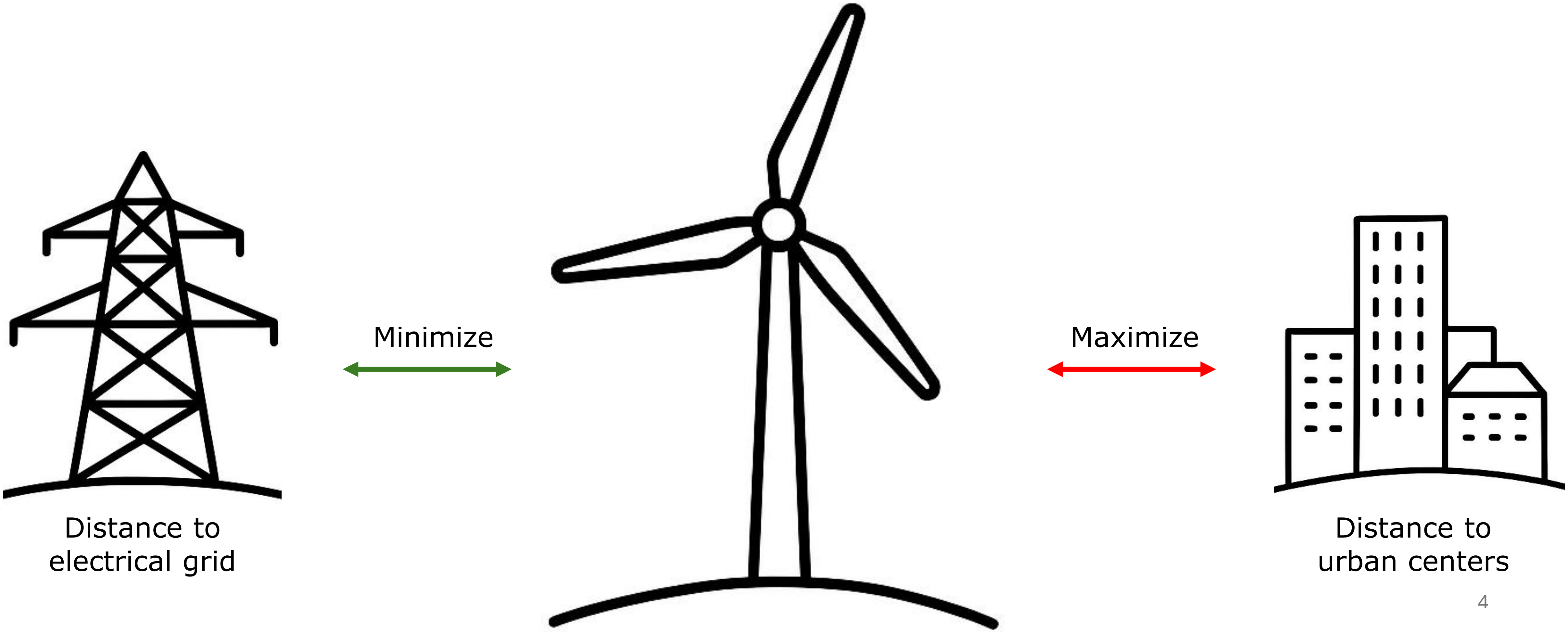
- Slope
- Distance to Urban Centers
- Distance to Transmission Lines and Substations
- Elevation
- Distance to Airports
- Distance to Roads
- Distance to Military Zones

Environmental & Ecological

- Distance to Protected Areas
- Flood Risk
- Distance to Migration Zones
- Distance to Agricultural Areas
- Distance to Water Bodies
- Landslide Risk

Problem Introduction

Variables are often **competing** in wind farm siting decisions



Problem Introduction

The field of Multi-Criteria Decision Making (**MCDM**) focuses on developing systematic approaches to rank alternatives. These approaches assign weights to decision factors **based on expert judgment**, capturing their relative importance

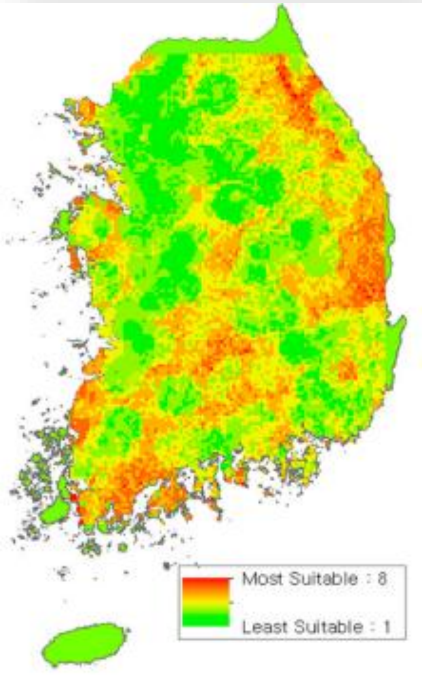
Common MCDM Methods include for example:

- Analytic Hierarchy Process (AHP)
- Fuzzy AHP
- Bayesian Best-Worst Method
- TOPSIS

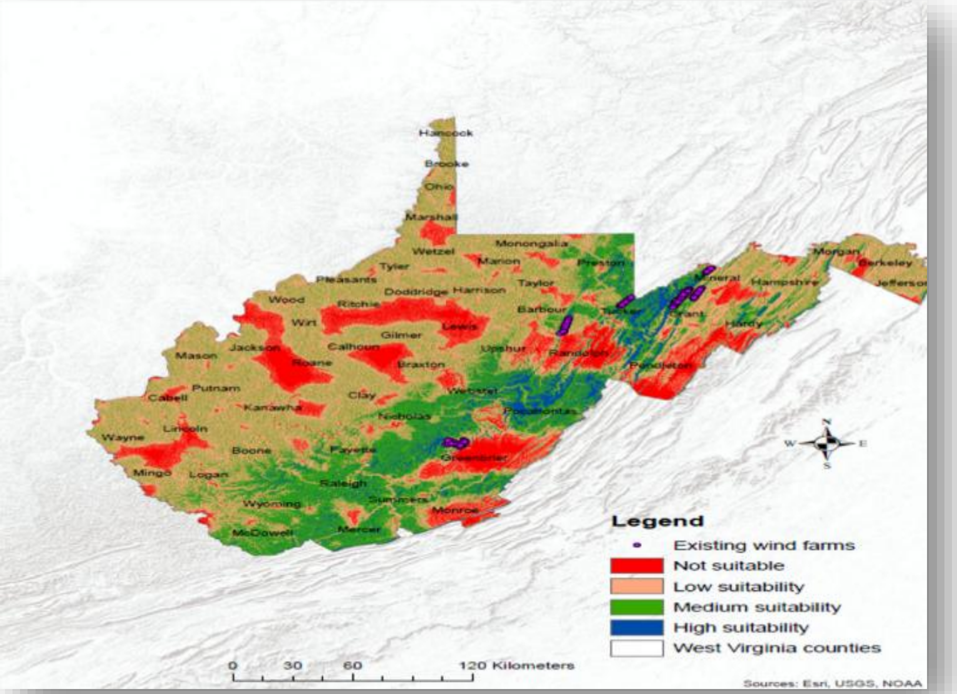
Factor	Weight
Wind speed	0.3982
Air density	0.1327
Slope	0.2151
Distance to electrical substations	0.1009
Distance to road network	0.0432
Distance to urban area	0.0432
Distance to transmission lines	0.0185
Distance to charging ports	0.0092
Vegetation coverage and land use	0.0390

Problem Introduction

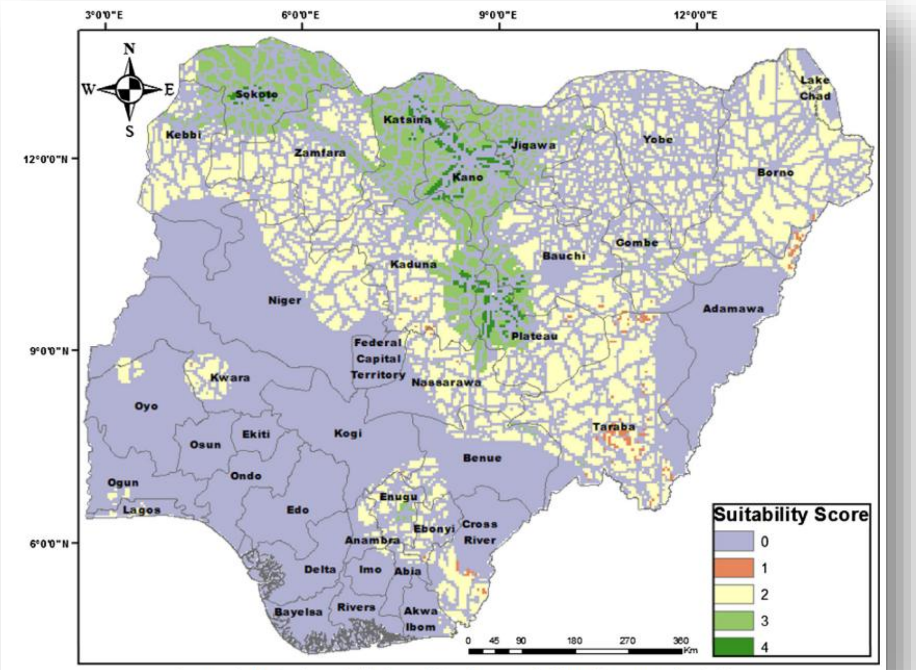
It's easy to visualize suitability maps, but it's hard to understand what was the **decision making process** behind them



(South Korea)



(West Virginia)

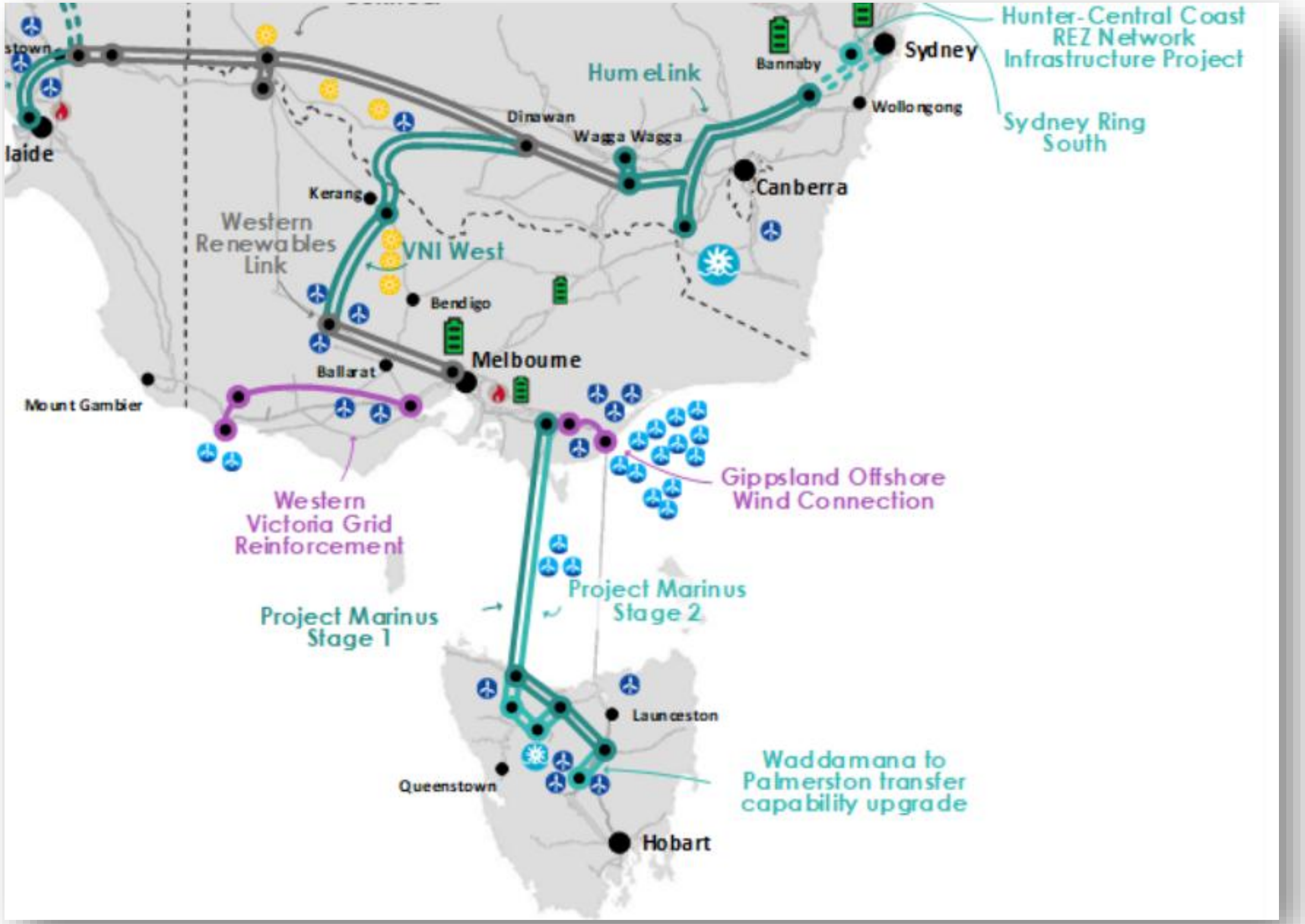


0 = not suitable, 1 = less suitable, 2 = suitable, 3 = very suitable and 4 = extremely suitable

(Nigeria)

Problem Introduction

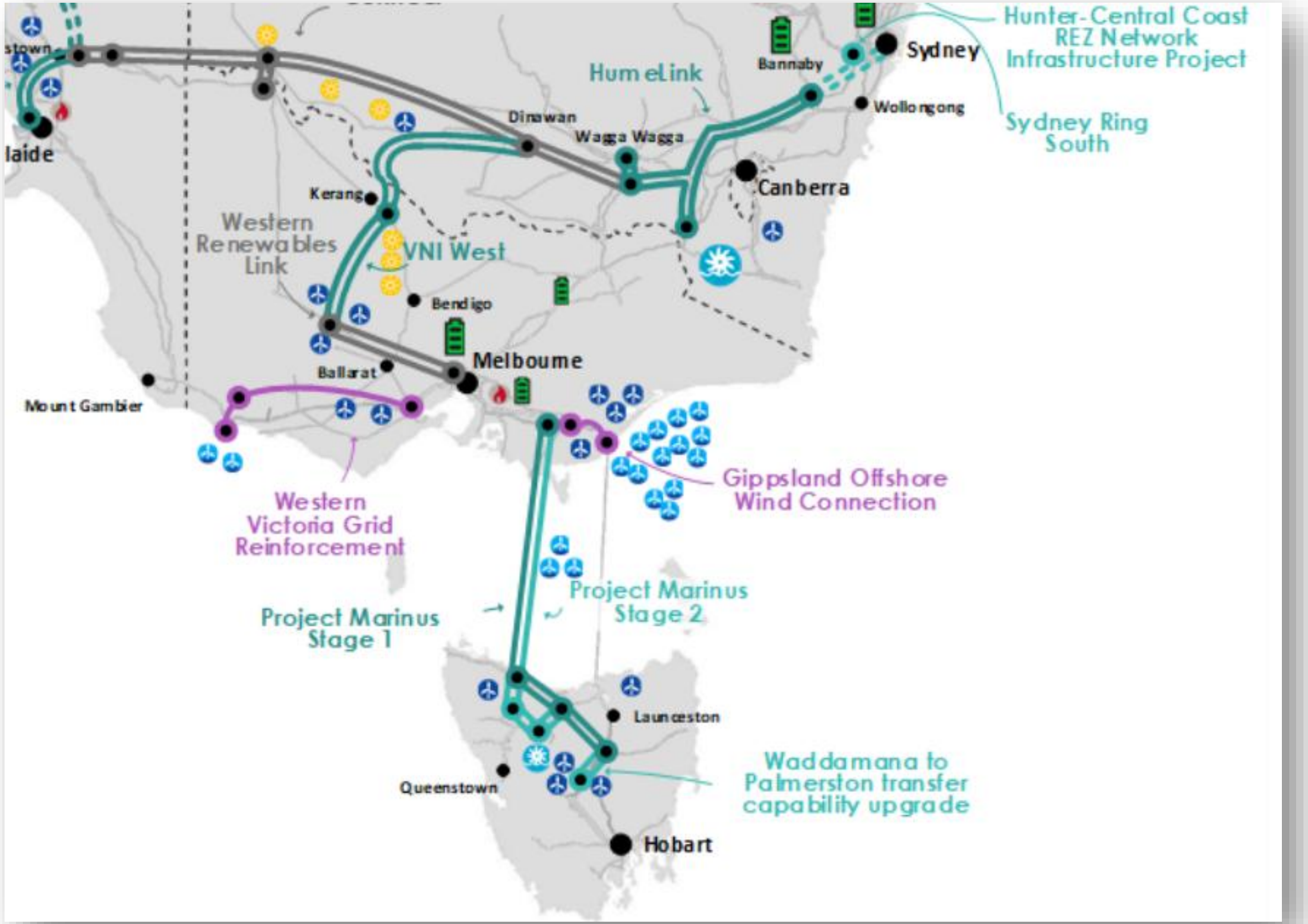
Communication Challenges



([Integrated System Plan 2024, Aemo])

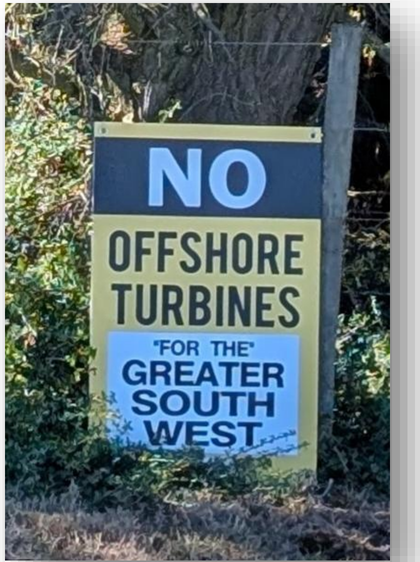
Problem Introduction

Communication Challenges



([Integrated System Plan 2024, Aemo]())

(Wollongong, May 2025)



(Warrnambool, Mar 2025)

(Canberra, Feb 2024)

Problem Introduction

Goal of the project

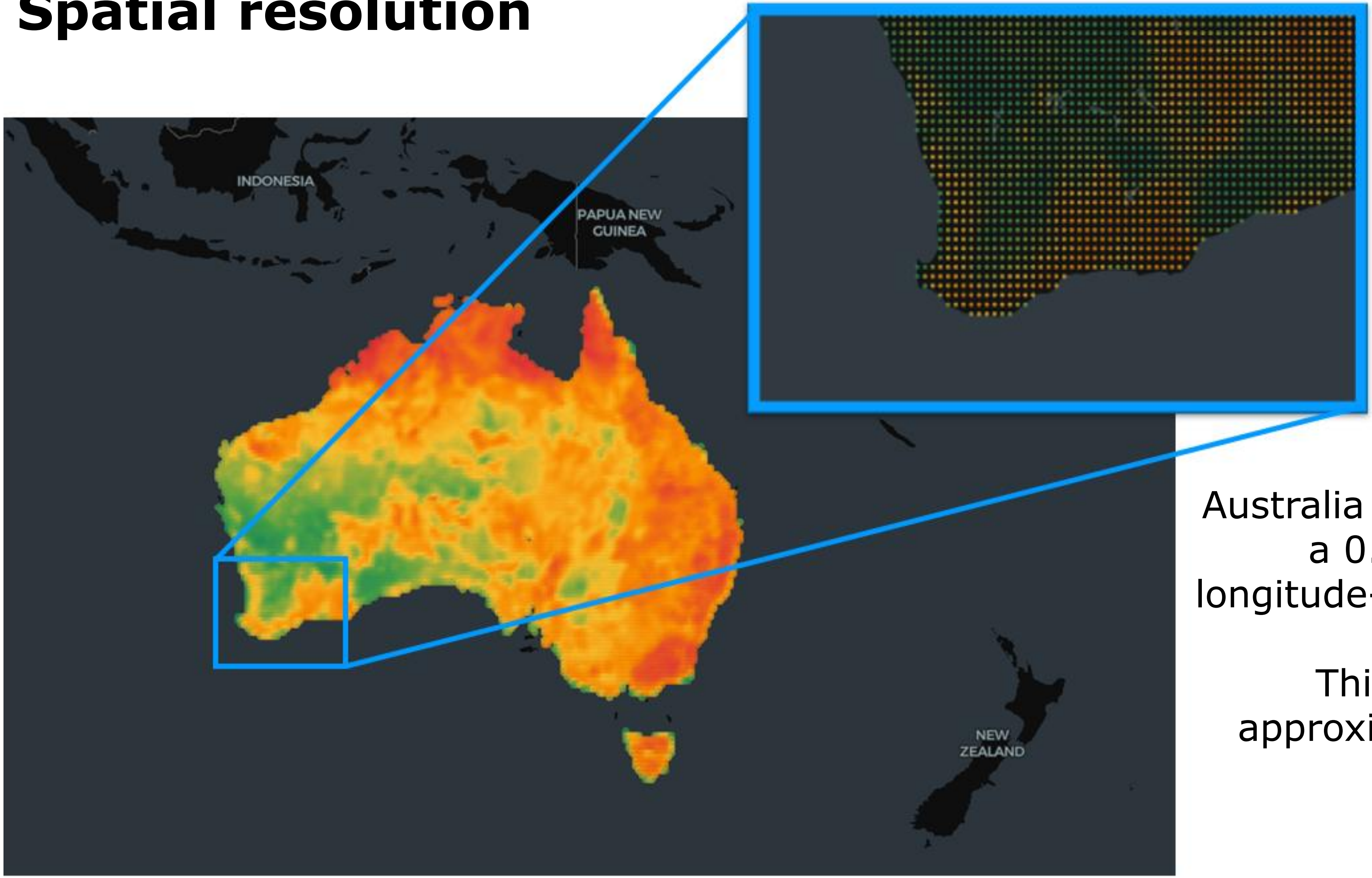
Build a site selection platform

That addresses the **specific** inefficiencies of site selection in Australia

That is **interactive** and **explainable**

Implementation

Spatial resolution



Australia was gridded using a 0.25° × 0.25° longitude-latitude resolution

This resulted in approximately **12k** grid points

Variables

For each point 5 variables were collected

- Distance to Electrical Grid
- Distance to Nature Land
- Solar Radiation (avg)
- Wind Capacity Factor (avg)
- Correlation with Existing Farms

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- **Wind Capacity Factor (avg)**
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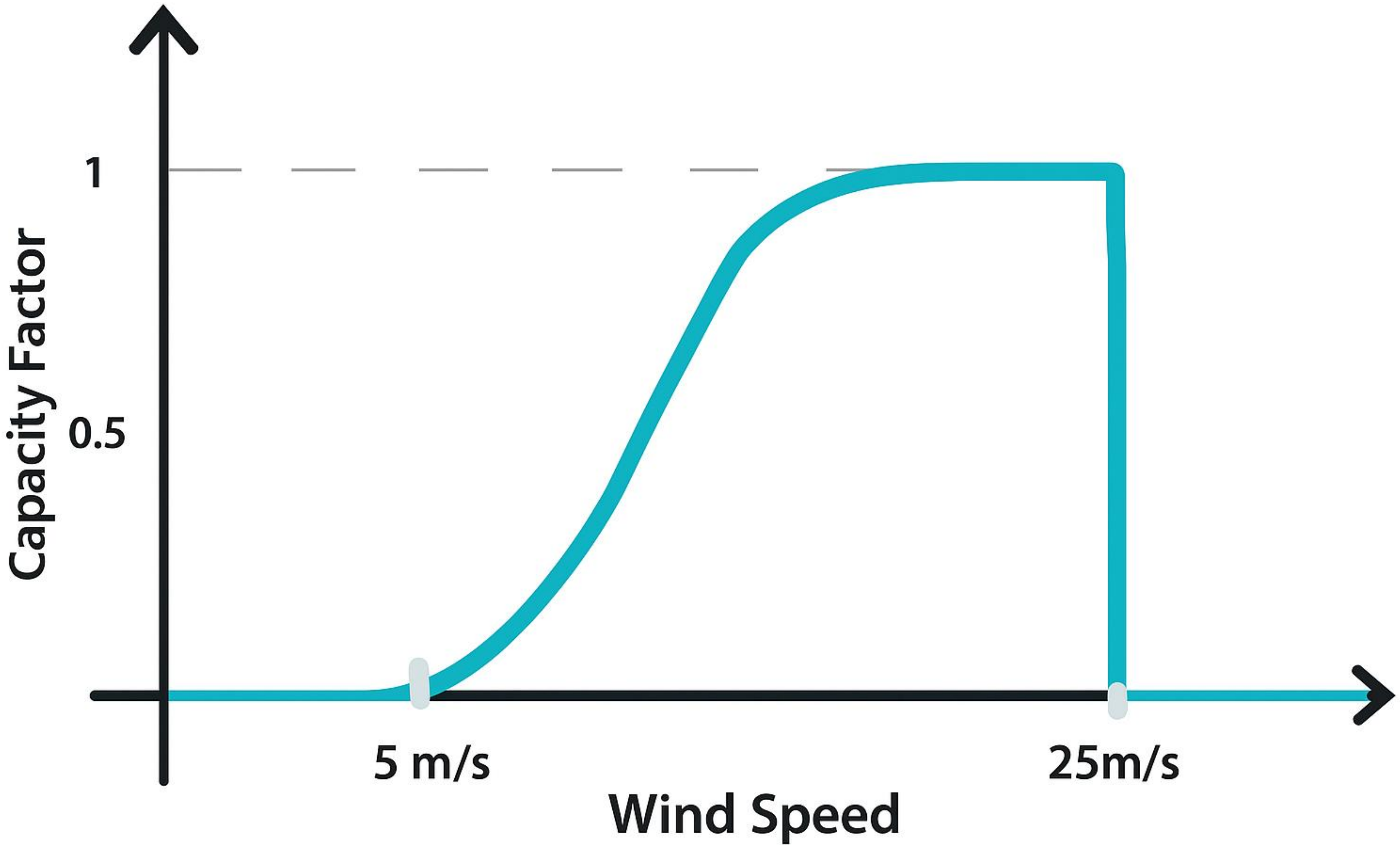
Wind Capacity Factor

Used as an indirect indicator of the wind resource at a site



Variables

For each point 5 variables were collected

- Distance to Electrical Grid
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- Avg Solar Radiation
- Wind Capacity Factor
- **Correlation with Existing Farms**

Goal of the project

Build a site selection platform

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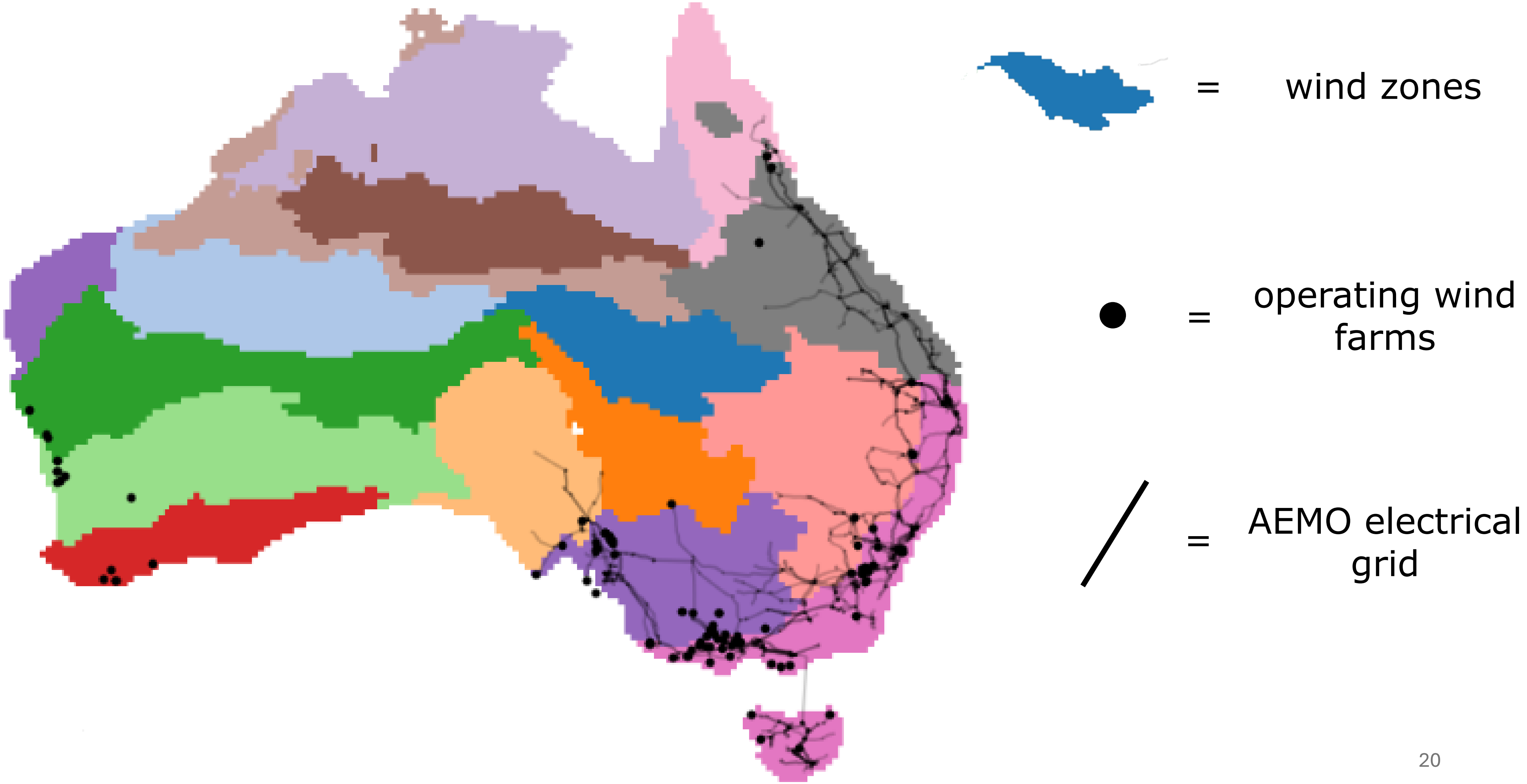
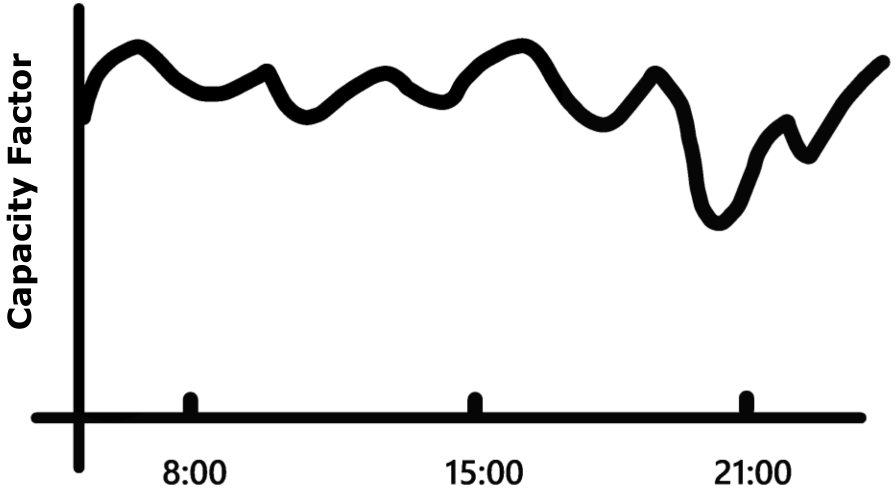
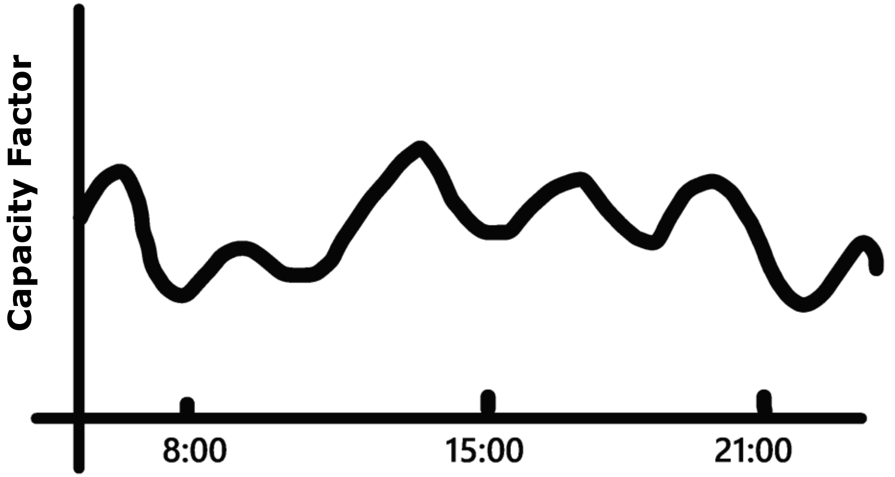
Spatial inefficiency



Correlation with Existing Farms



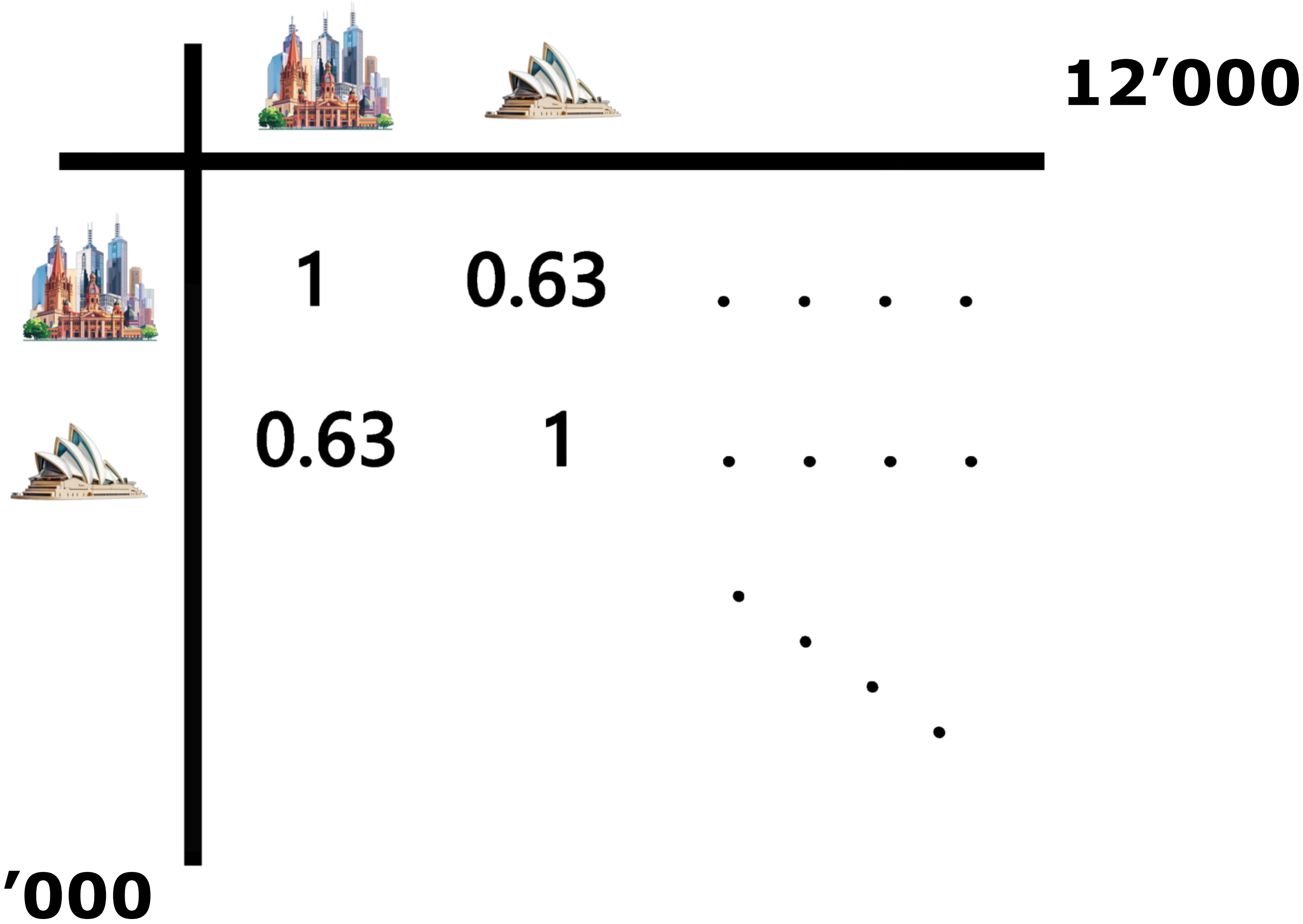


Corr: 0.63

Correlation with Existing Farms

		
	1	0.63
	0.63	1

Correlation with Existing Farms



Correlation with Existing Farms

					
	1	0.63	. . . .	0.82	0.58
	0.63	1	. . . .	0.21	0.09
			. . . .		

Correlation with Existing Farms

						
	1	0.63	. . . .	0.82	0.58	0.70
	0.63	1	. . . .	0.21	0.09	0.15

Variables

For each point 5 variables were collected

- Distance to Electrical Grid
- Distance to Nature Land
- Solar Radiation (avg)
- Wind Capacity Factor (avg)
- Correlation with Existing Farms

Suitability Index

How can we combine these values to form a suitability index?

	Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
1.	3km	278 W/m^2	39km	0.74	0.53
2.	456km	293 W/m^2	211km	0.51	0.37
	…	…	…	…	…
12’000.	91km	315 W/m^2	69km	0.14	0.19

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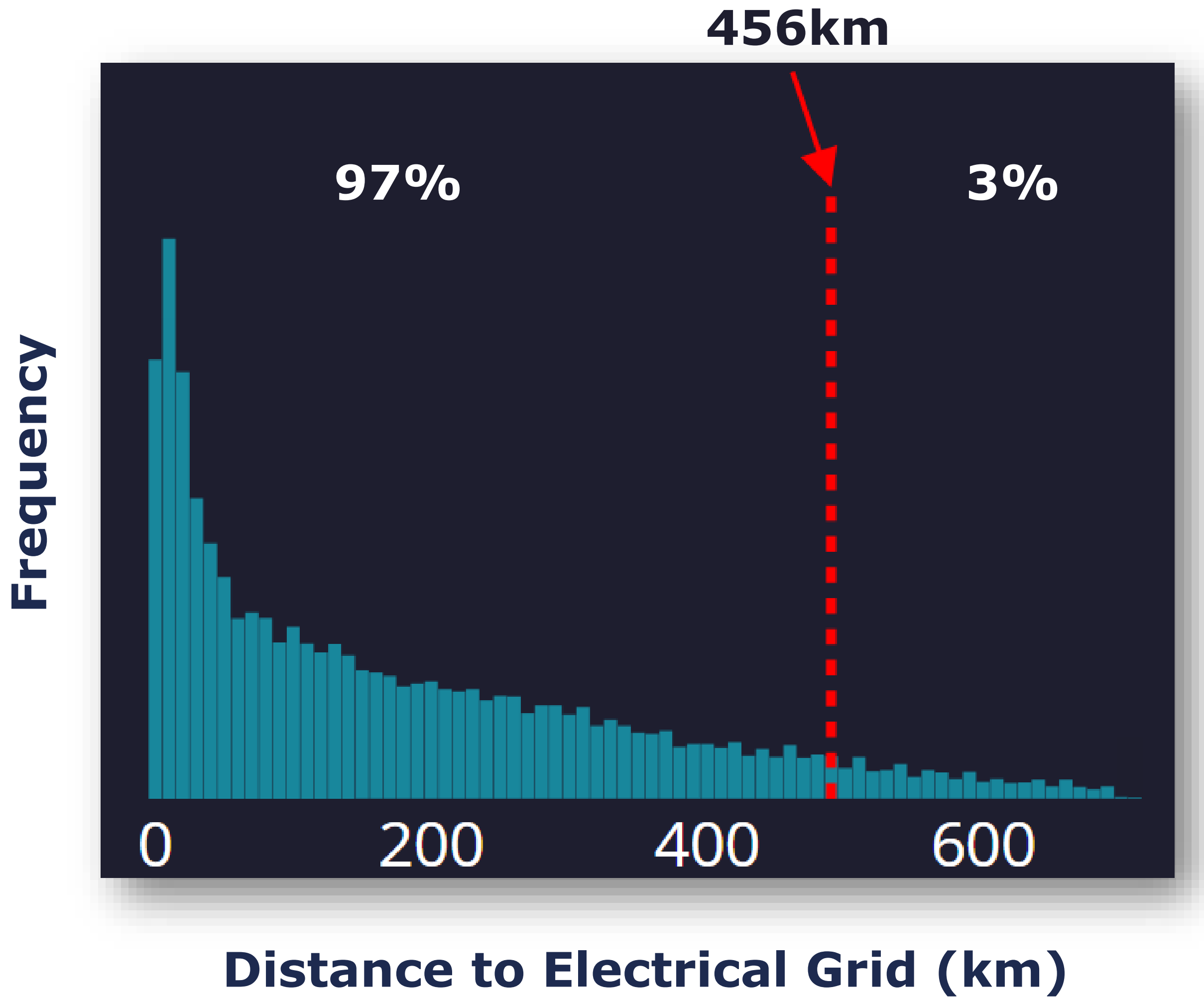
Suitability Index



Distance to Electrical Grid Score

3

Suitability Index

How can we combine these values to form a suitability index?

	Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
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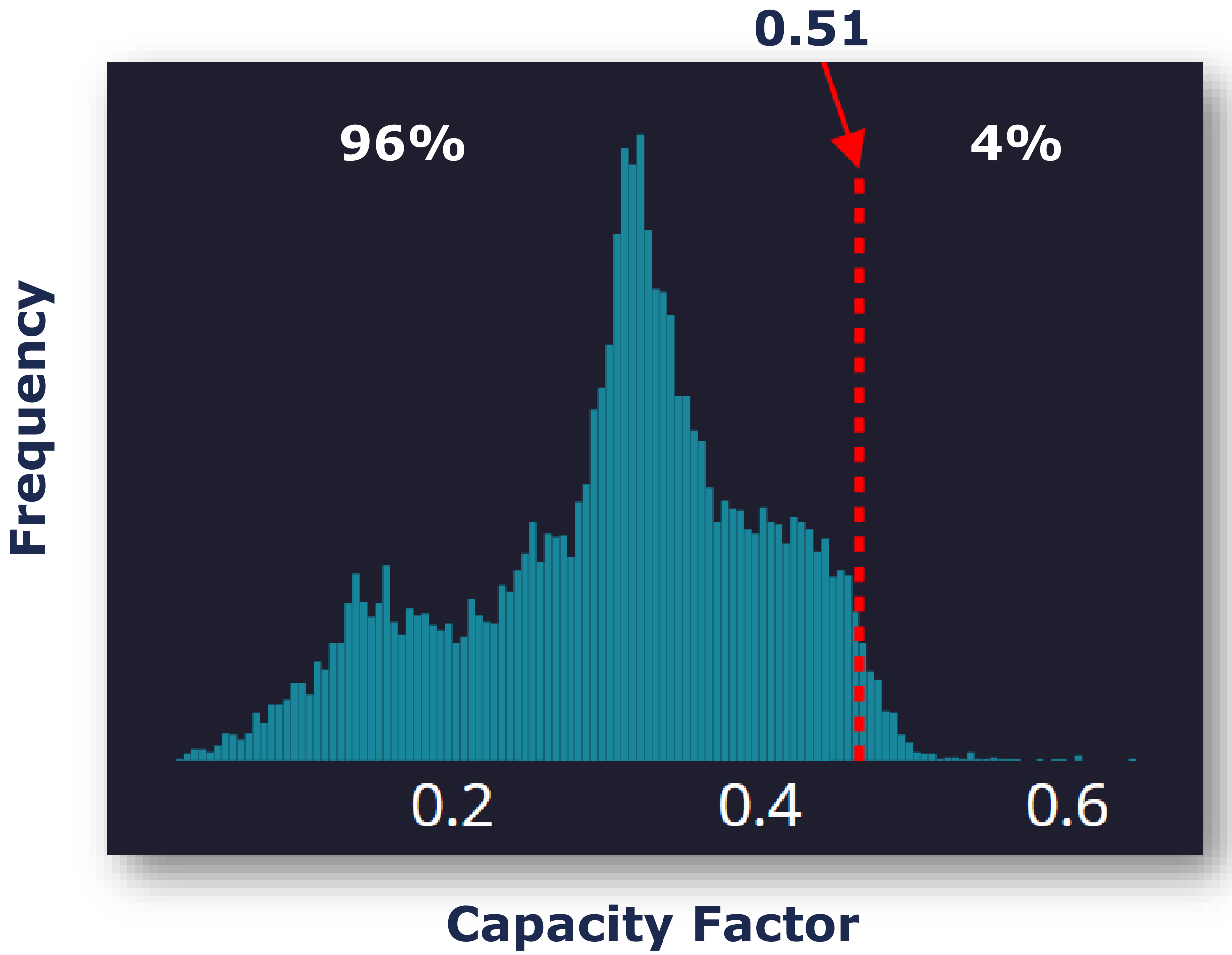
12’000.	91km	315 W/m²	69km	0.14	0.19

Suitability Index



Capacity Factor Score

96

Suitability Index

How can we combine these values to form a suitability index?

	Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
1.	3km	278 W/m^2	39km	0.74	0.53
2.	456km	293 W/m^2	211km	0.51	0.37
	…	…	…	…	…
12’000.					

Suitability Index

	Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
1.	7	57	97	87	13
2.	3	63	17	96	82
	…	…	…	…	…
12’000.					

Suitability Index

Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
7	57	97	87	13

Suitability Index

Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
W1* 7	W2* 57	W3* 97	W4* 87	W4* 13

Suitability Index

Distance to Electrical Grid		Avg Solar Radiation		Distance to Nature Land		Wind Capacity Factor		Correlation with Existing Farms
W1* 7	+	W2* 57	+	W3* 97	+	W4* 87	+	W4*13

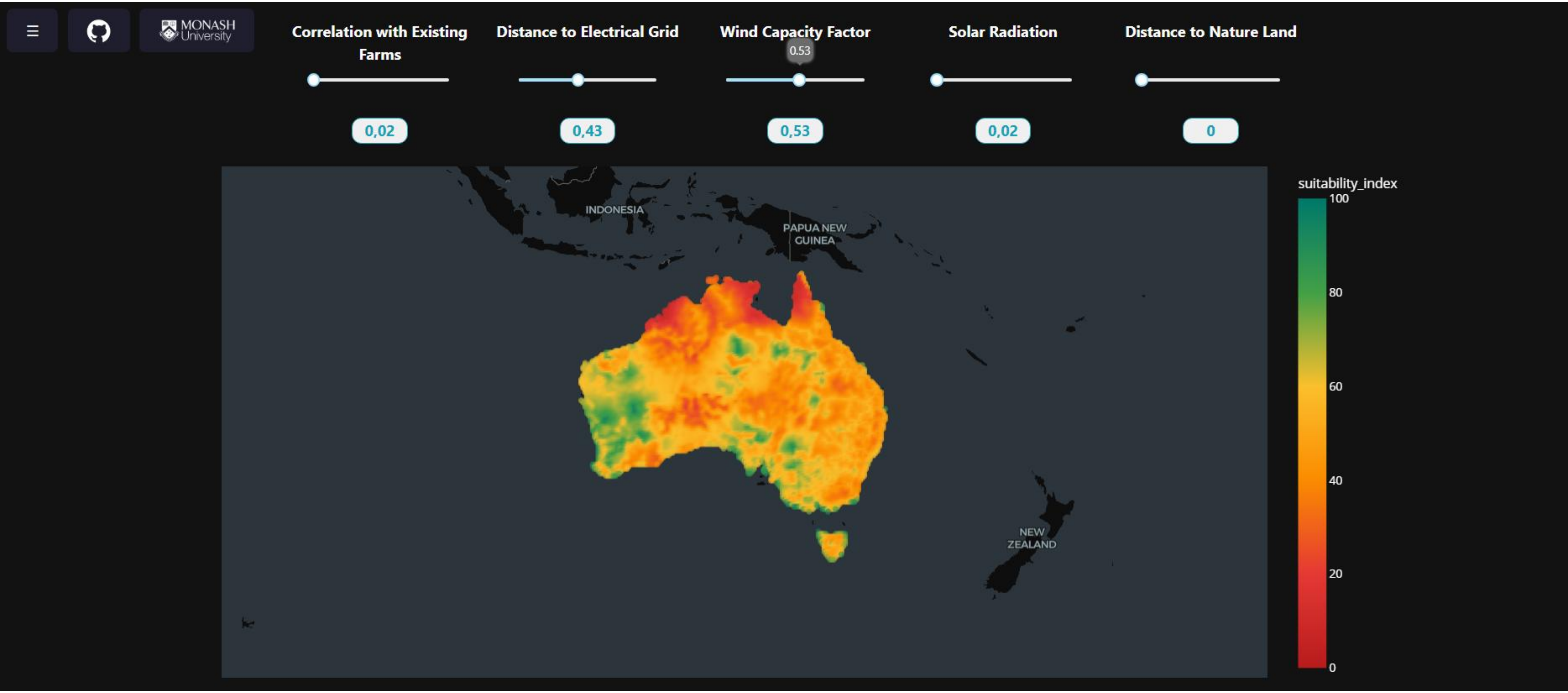
Suitability Index

Distance to Electrical Grid	Avg Solar Radiation	Distance to Nature Land	Wind Capacity Factor	Correlation with Existing Farms
W1* 7	W2* 57	W3* 97	W4* 87	W4*13

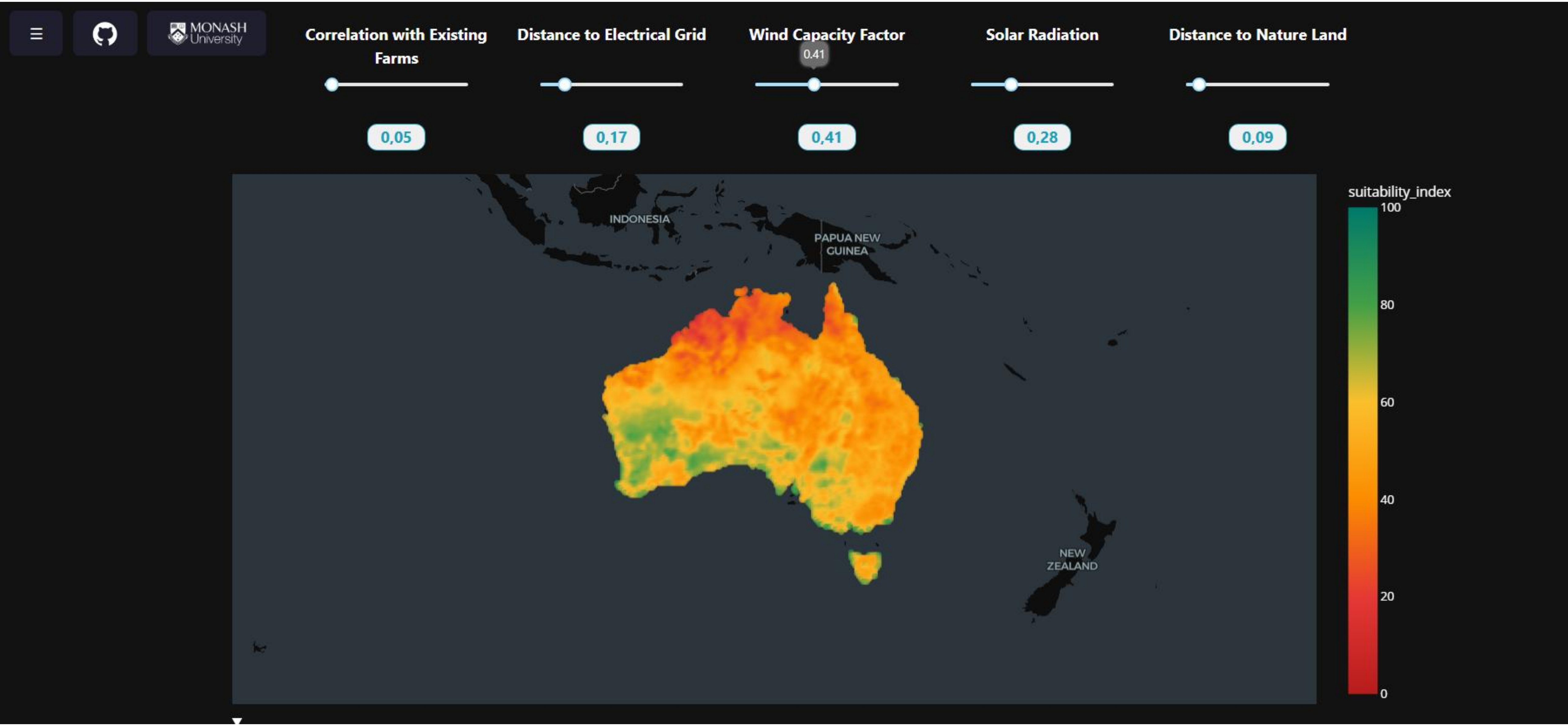
W1* 7 + W2* 57 + W3* 97 + W4* 87 + W4*13

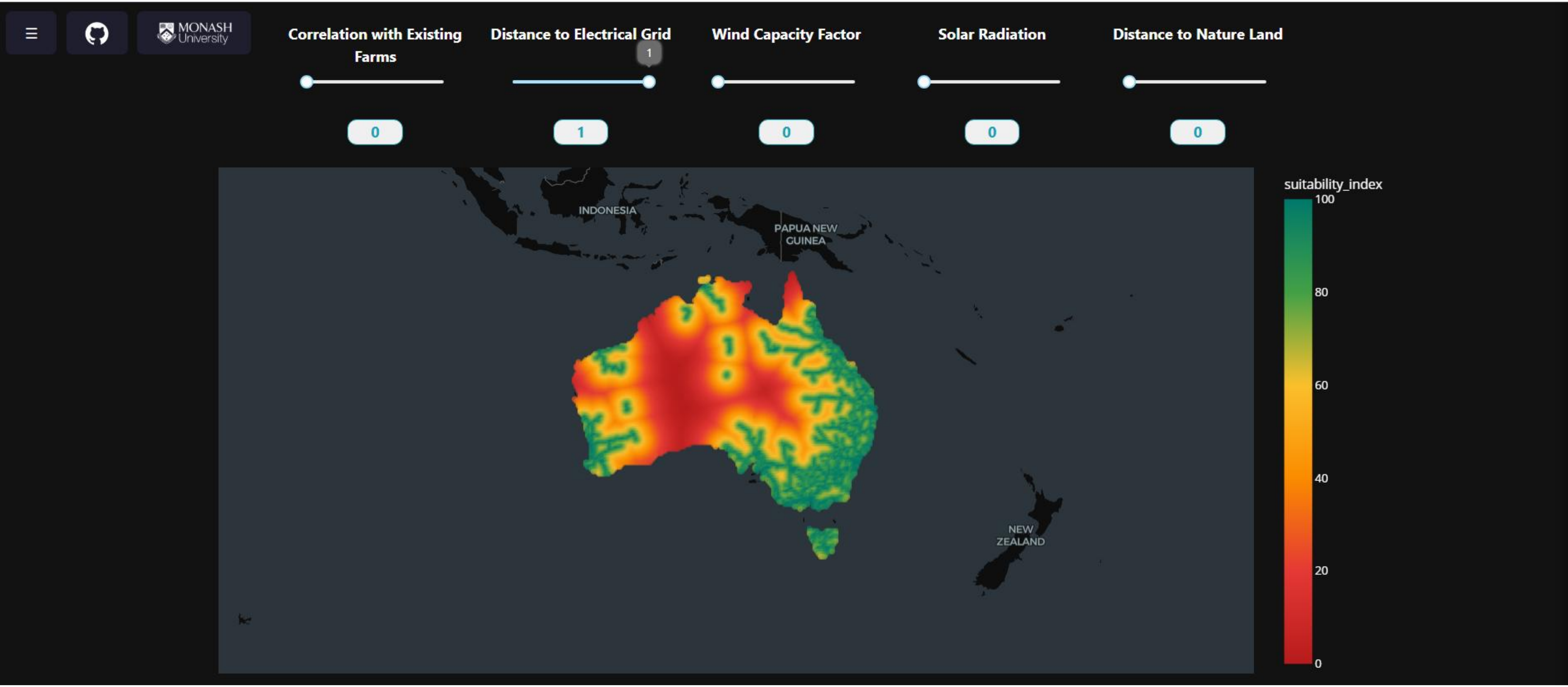
Suitability Index

77

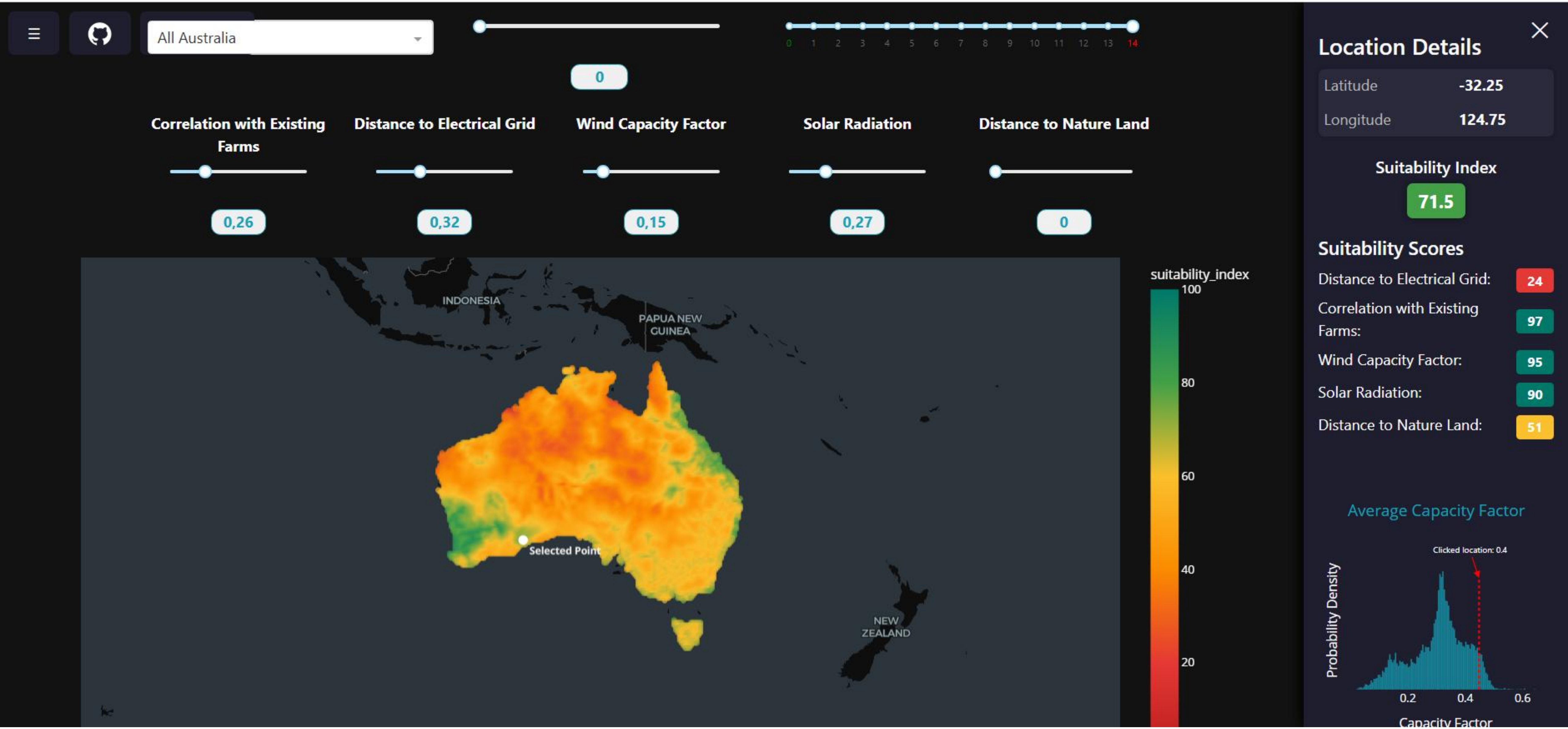


MONASH University
Correlation with Existing Farms
Distance to Electrical Grid
Wind Capacity Factor
Solar Radiation
Distance to Nature Land
0.53
0,02
0,43
0,53
0,02
0
INDONESIA
PAPUA NEW GUINEA
NEW ZEALAND
suitability_index
100
80
60
40
20
0

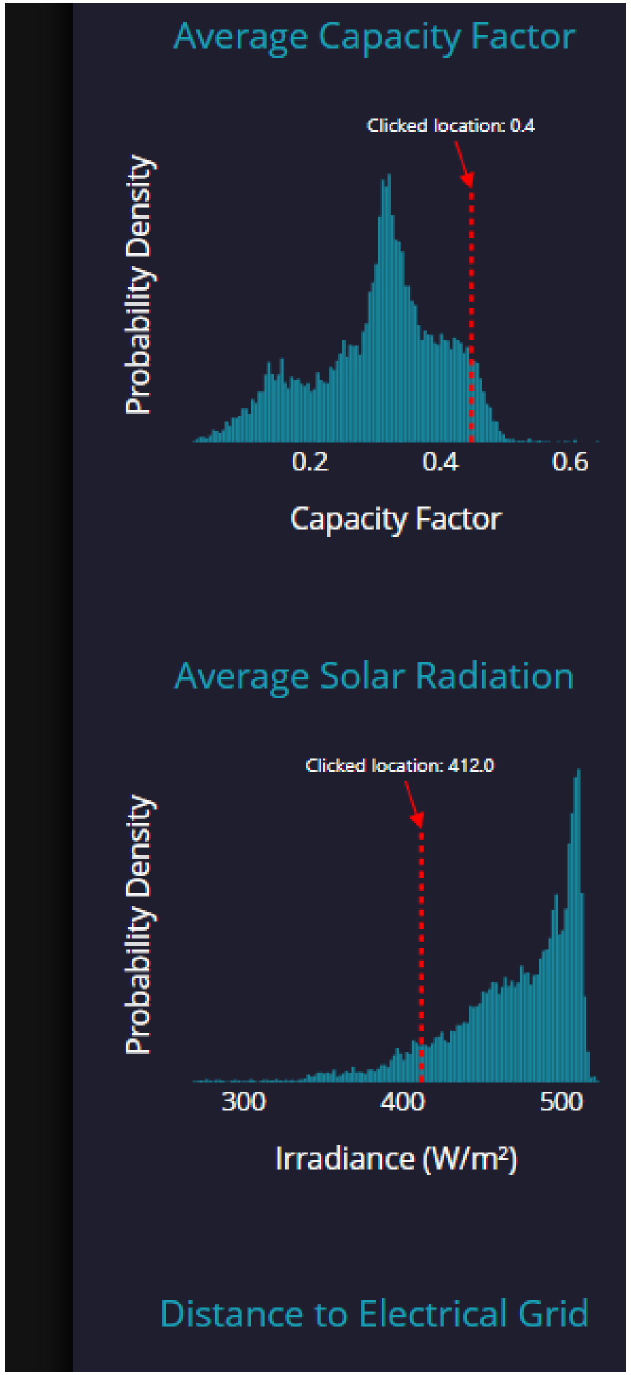

MONASH University
Correlation with Existing Farms
Distance to Electrical Grid
Wind Capacity Factor
0.41
Solar Radiation
Distance to Nature Land
0,05
0,17
0,41
0,28
0,09
INDONESIA
PAPUA NEW GUINEA
NEW ZEALAND
suitability_index
100
80
60
40
20
0



MONASH University
Correlation with Existing Farms
Distance to Electrical Grid
Wind Capacity Factor
Solar Radiation
Distance to Nature Land
0
1
0
0
0
INDONESIA
PAPUA NEW GUINEA
NEW ZEALAND
suitability_index
100
80
60
40
20
0



All Australia
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Correlation with Existing Farms
0,26
Distance to Electrical Grid
0,32
Wind Capacity Factor
0,15
Solar Radiation
0,27
Distance to Nature Land
0
INDONESIA
PAPUA NEW GUINEA
NEW ZEALAND
Selected Point
suitability_index
100
80
60
40
20
Location Details
Latitude -32.25
Longitude 124.75
Suitability Index
71.5
Suitability Scores
Distance to Electrical Grid: 24
Correlation with Existing Farms: 97
Wind Capacity Factor: 95
Solar Radiation: 90
Distance to Nature Land: 51
Average Capacity Factor
Clicked location: 0.4
Probability Density
0.2
0.4
0.6
Capacity Factor


Average Capacity Factor
Clicked location: 0.4
Probability Density
0.2
0.4
0.6
Capacity Factor
Average Solar Radiation
Clicked location: 412.0
Probability Density
300
400
500
Irradiance (W/m²)
Distance to Electrical Grid

Other features

Other features

Pareto optimization

	Location A	dominates	**Location B**
Distance to Electrical Grid	**63**		**63**
Avg Solar Radiation	**17**		**17**
Distance to Nature Land	**97**		**97**
Wind Capacity Factor	**31**		**31**
Correlation with Existing Farms	**85**		**69**

Other features

Pareto optimization

	Location A	dominates	Location B
Distance to Electrical Grid	63		63
Avg Solar Radiation	17		17
Distance to Nature Land	97		97
Wind Capacity Factor	31		31
Correlation with Existing Farms	85		69

Location A is no worse than Location B in all objectives

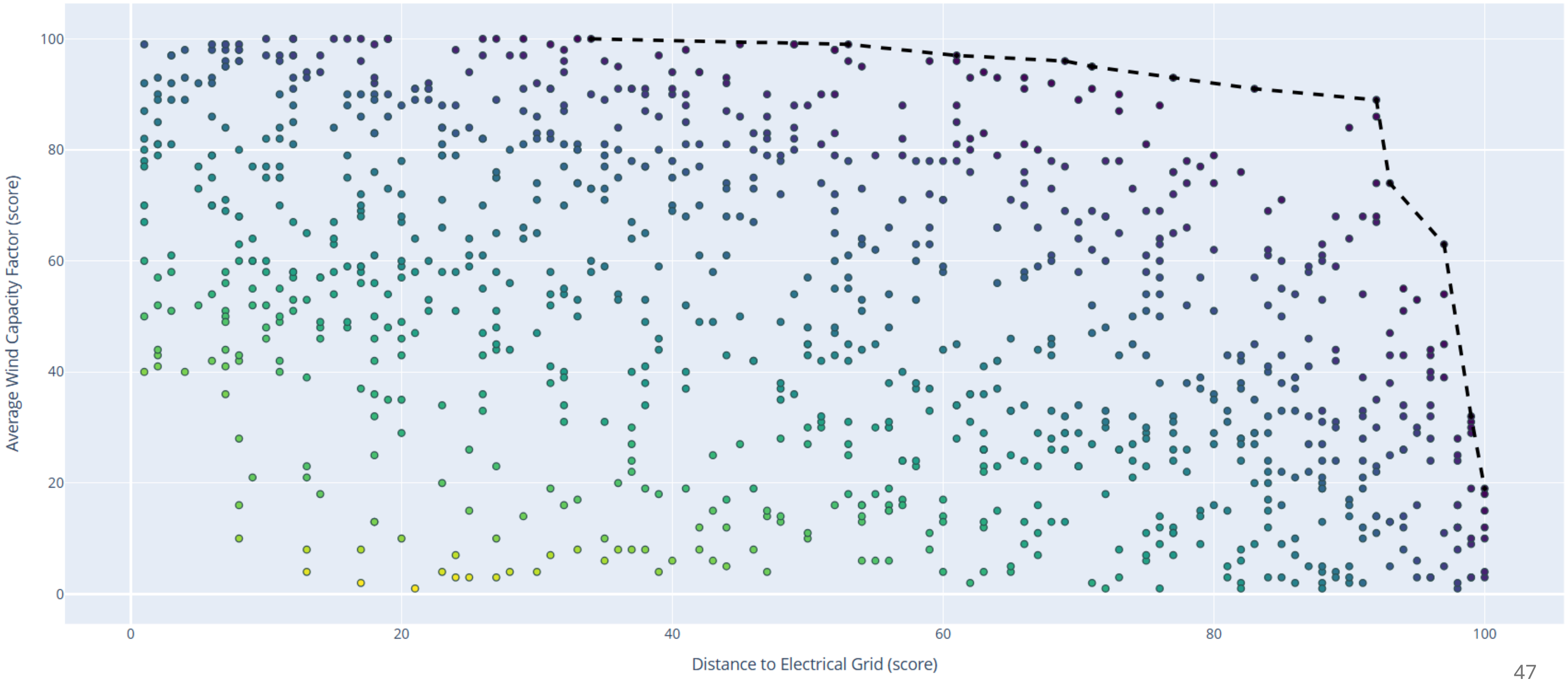
Other features

Pareto optimization

	Location A	dominates	Location B	
Distance to Electrical Grid	63		63	Location A is no worse than Location B in all objectives
Avg Solar Radiation	17		17	
Distance to Nature Land	97		97	
Wind Capacity Factor	31		31	
Correlation with Existing Farms	85		69	Location A is strictly better than Location B in at least one objective

Extra

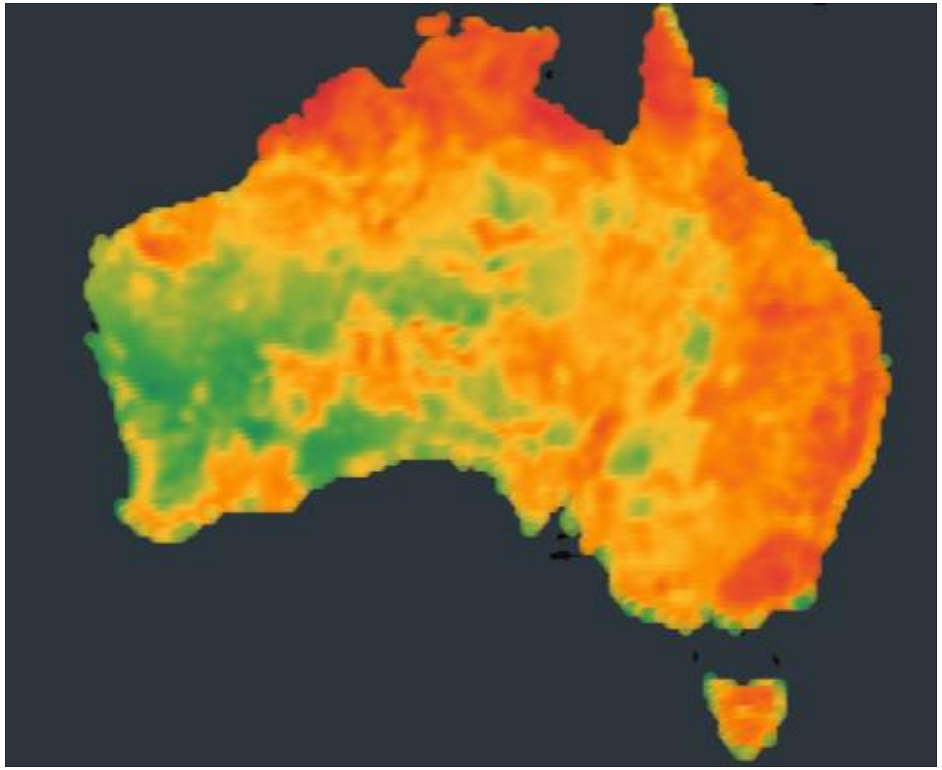
Pareto optimization



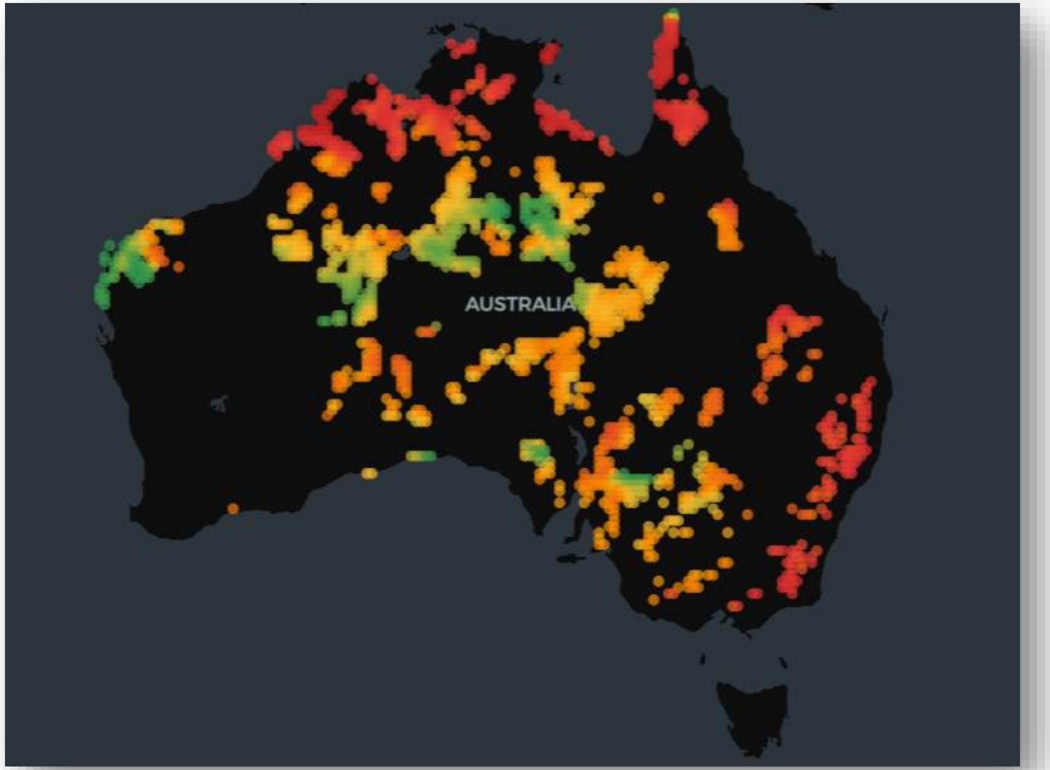
Other features

Pareto optimization

All locations

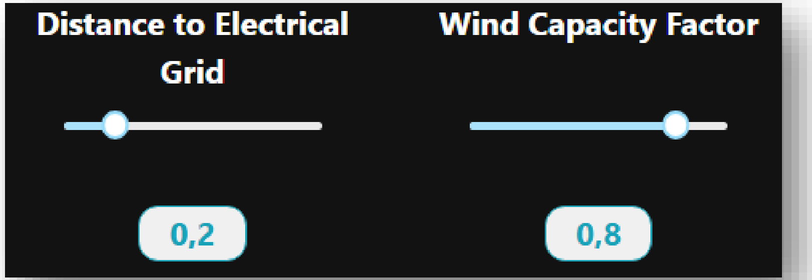


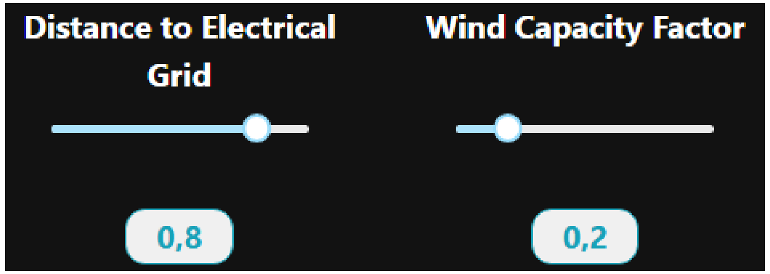
Pareto optimal locations

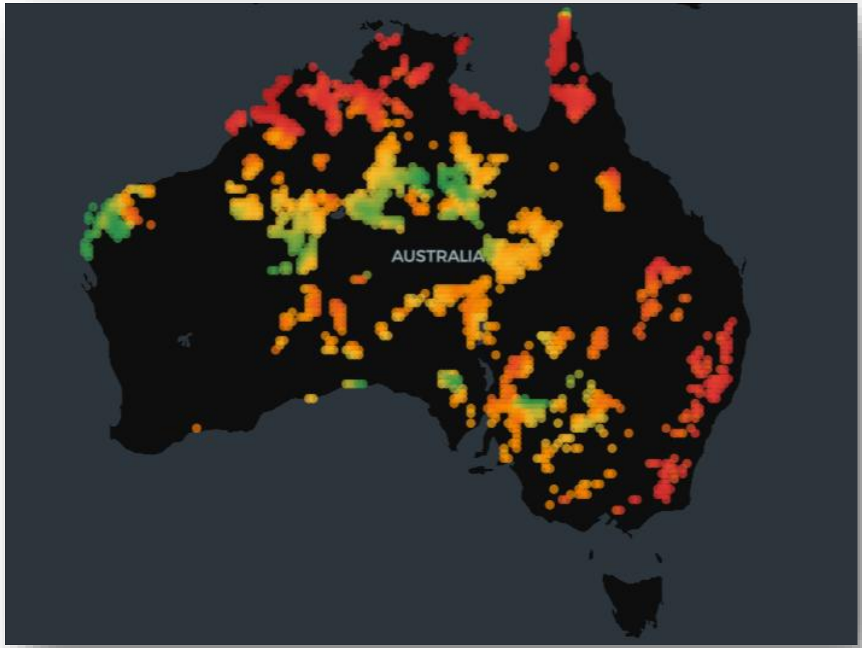


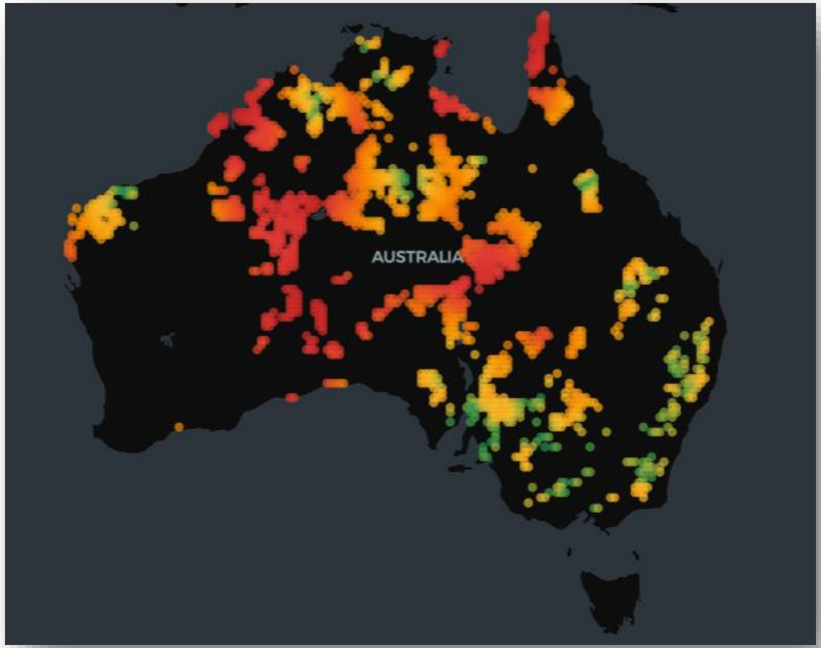
Other features

Pareto optimization









Summary

A site selection platform

- That is **interactive** and **explainable,** it allows to create **"what if"** scenarios

- Includes a **novel** variable to address inefficiencies in the current configuration

- This prototype demonstrates the **concept**, with potential for further refinement through additional layers

UNIVERSITÀ DEGLI STUDI FIRENZE

Da un secolo, oltre.

MONASH University

HR EXCELLENCE IN RESEARCH

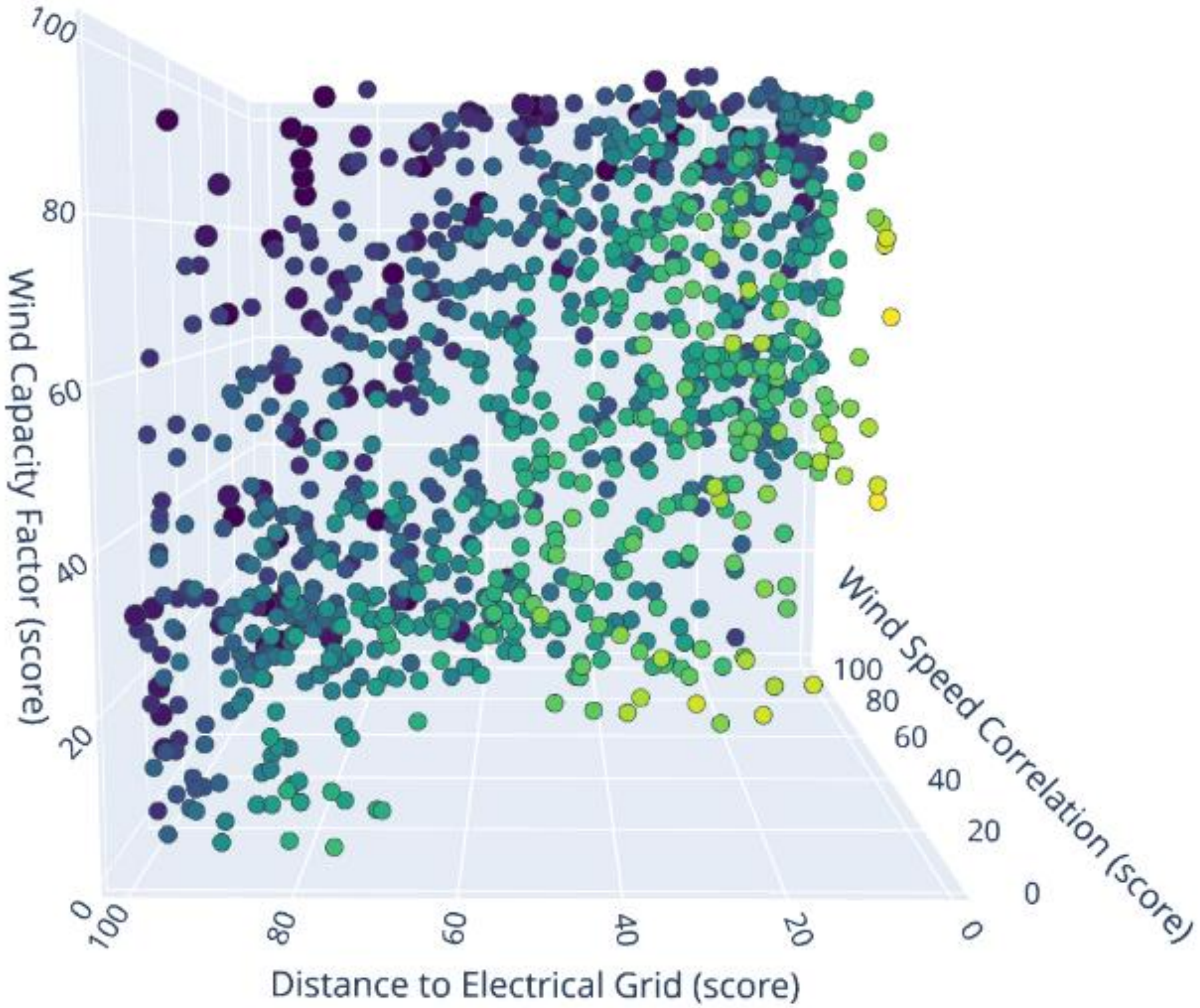
A platform for supporting onshore windfarm siting in Australia

Supervisors: Simone Marinai, Tim Dwyer

Extra

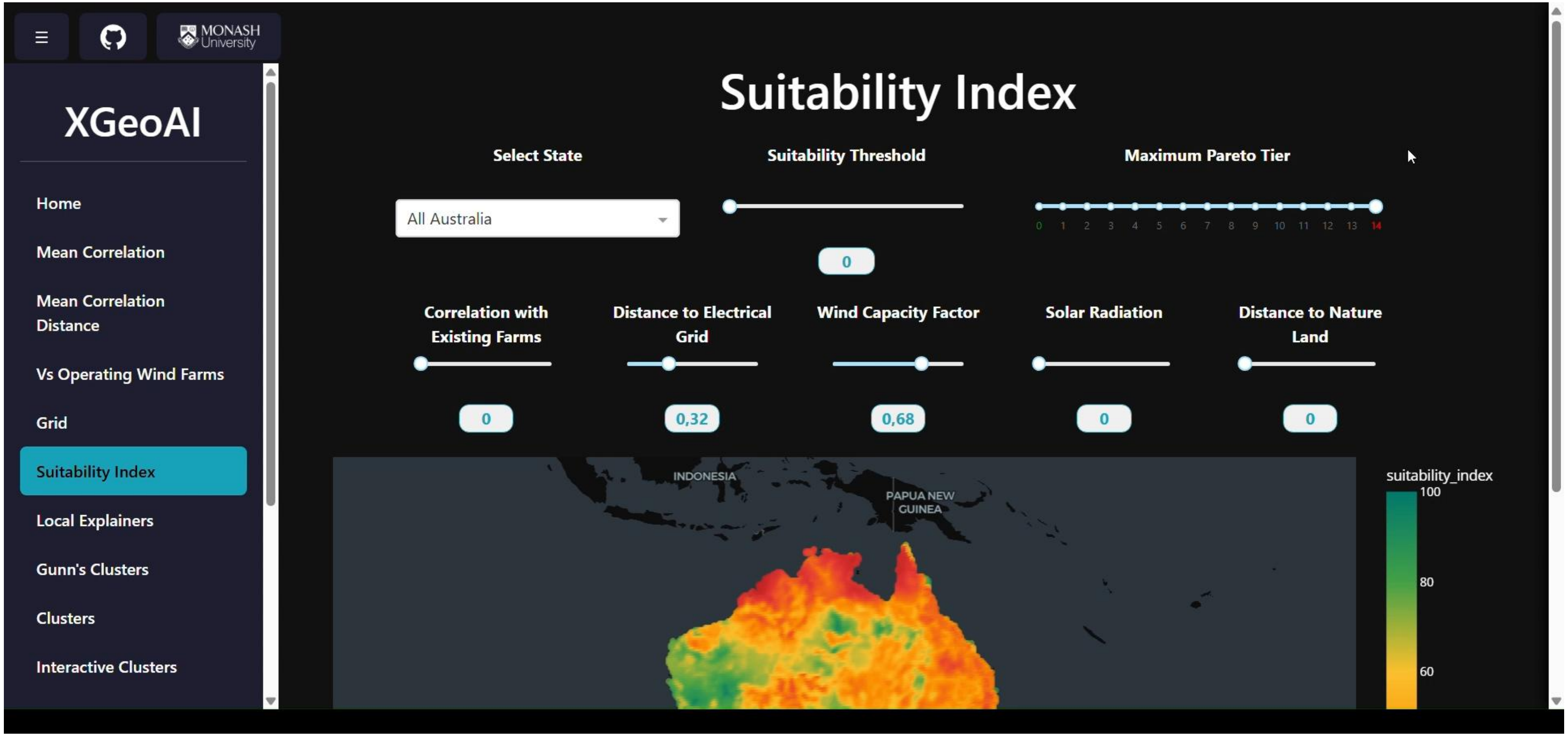
Extra

Pareto optimization



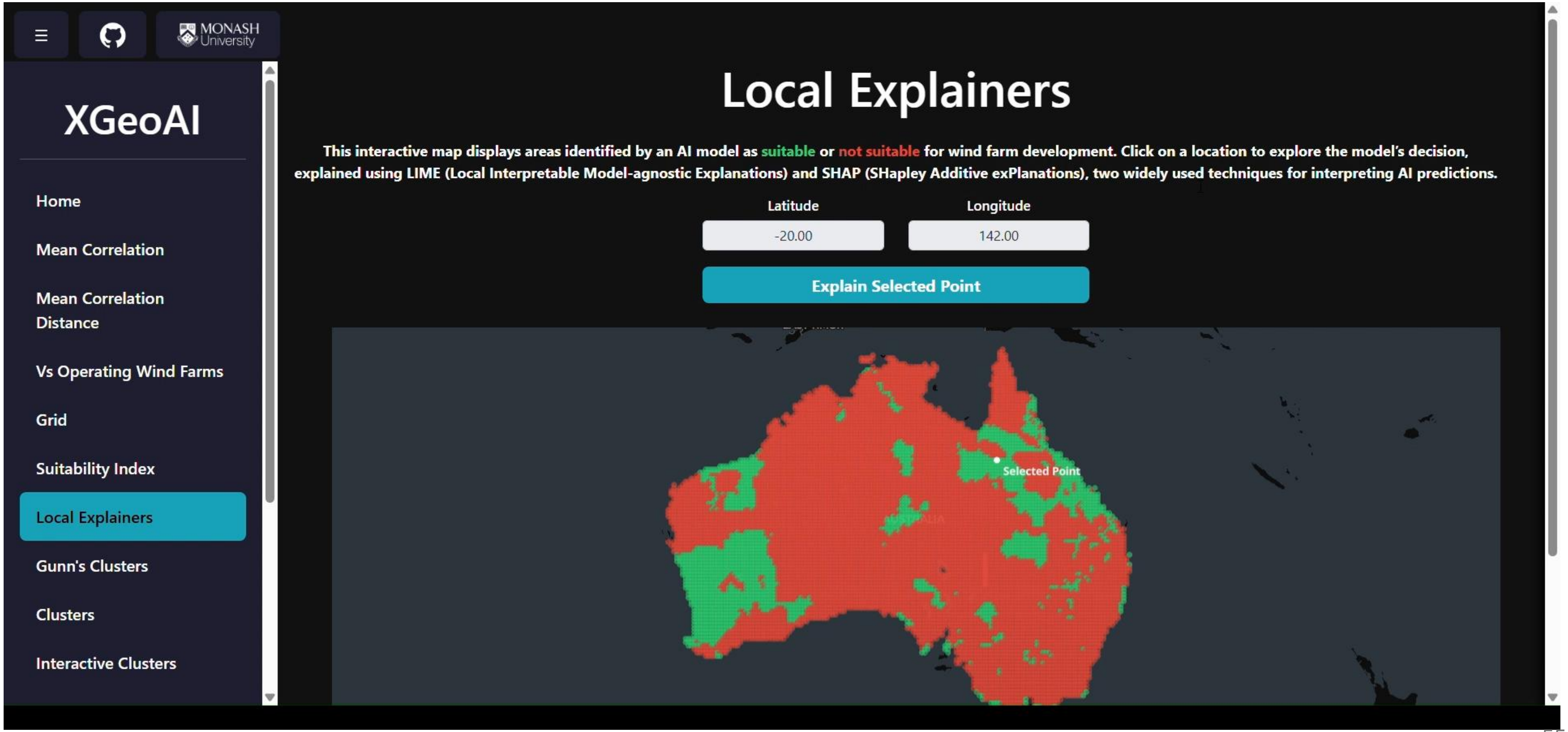
Extra

Pareto optimization



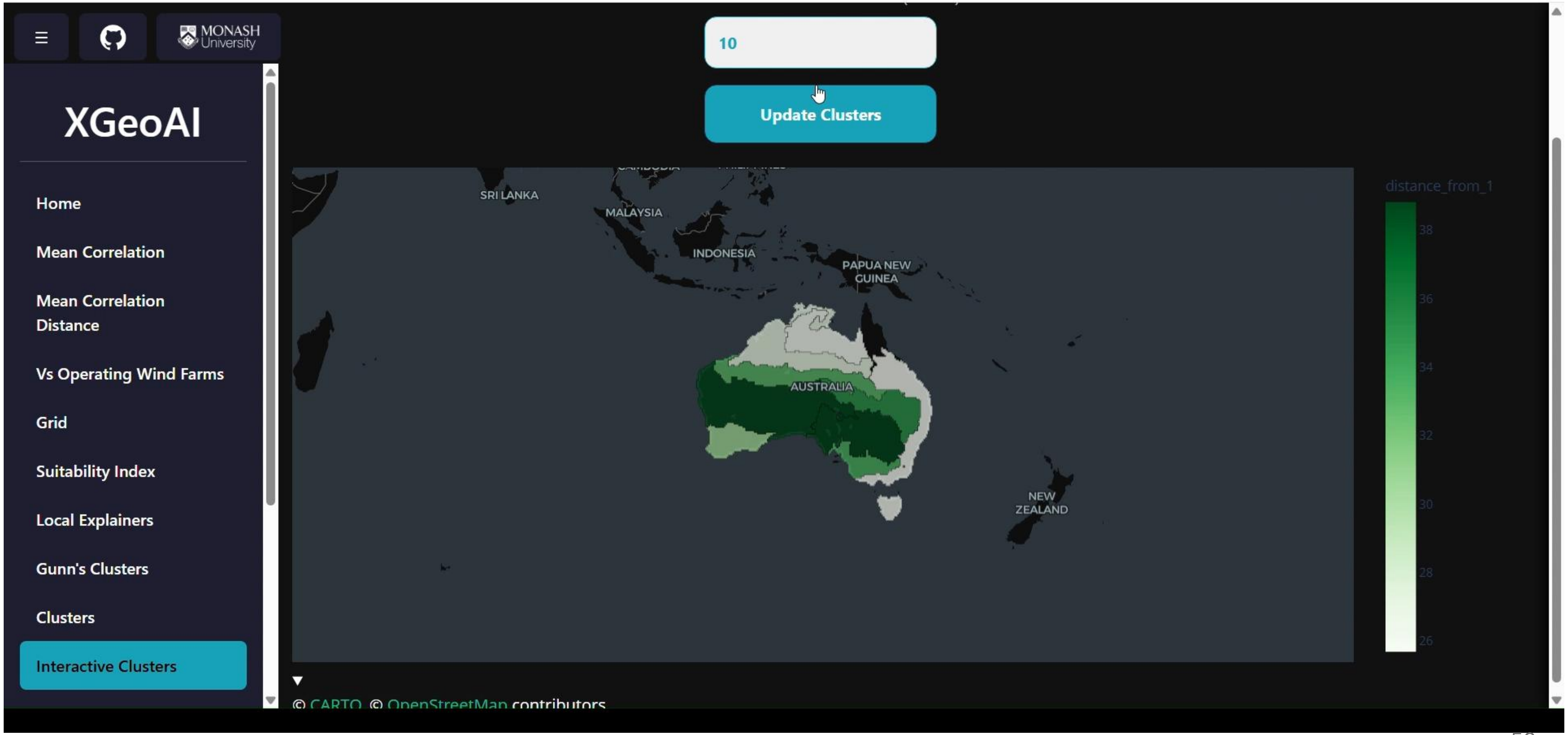
Extra

Local explainers



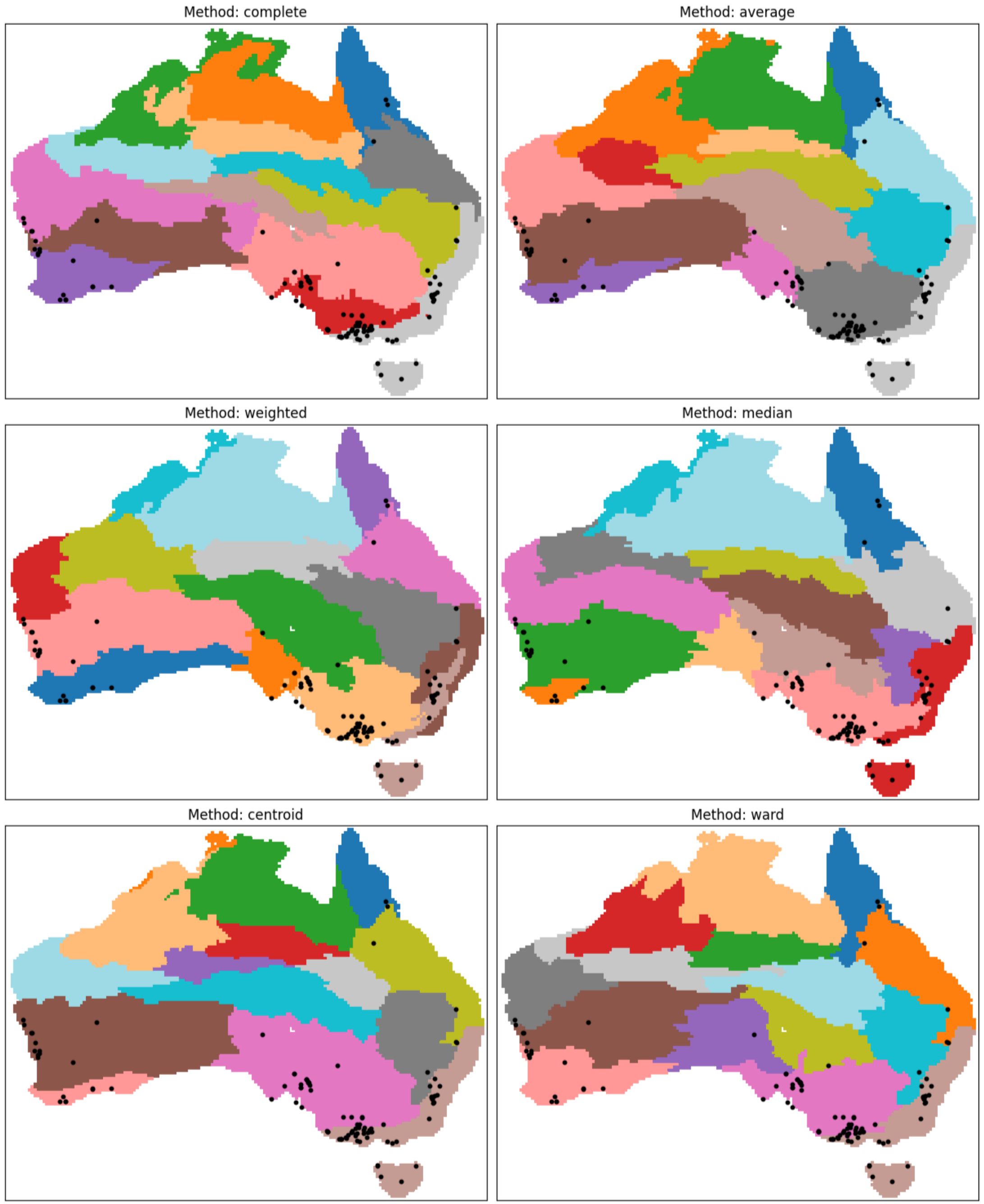
Extra

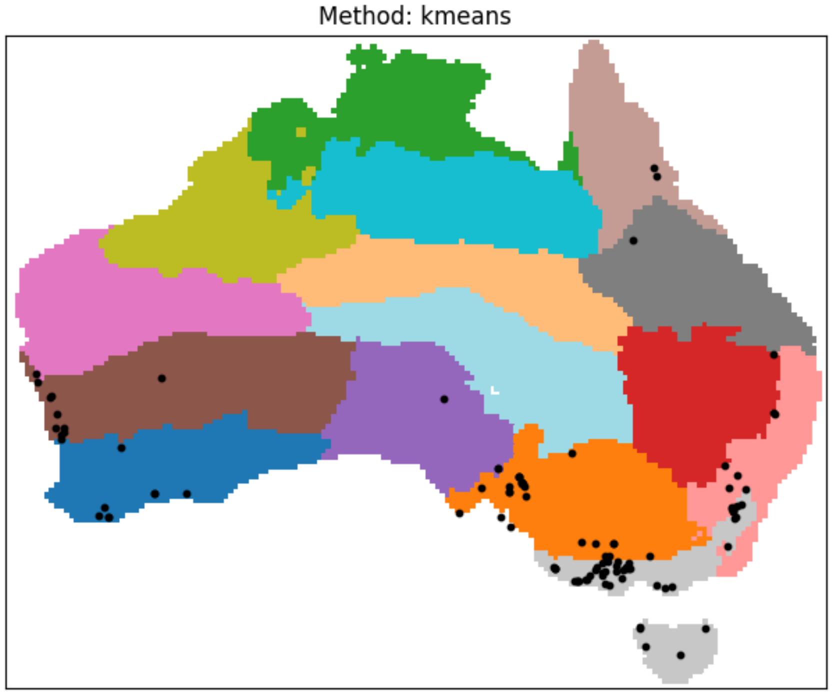
Interactive clusters



Extra

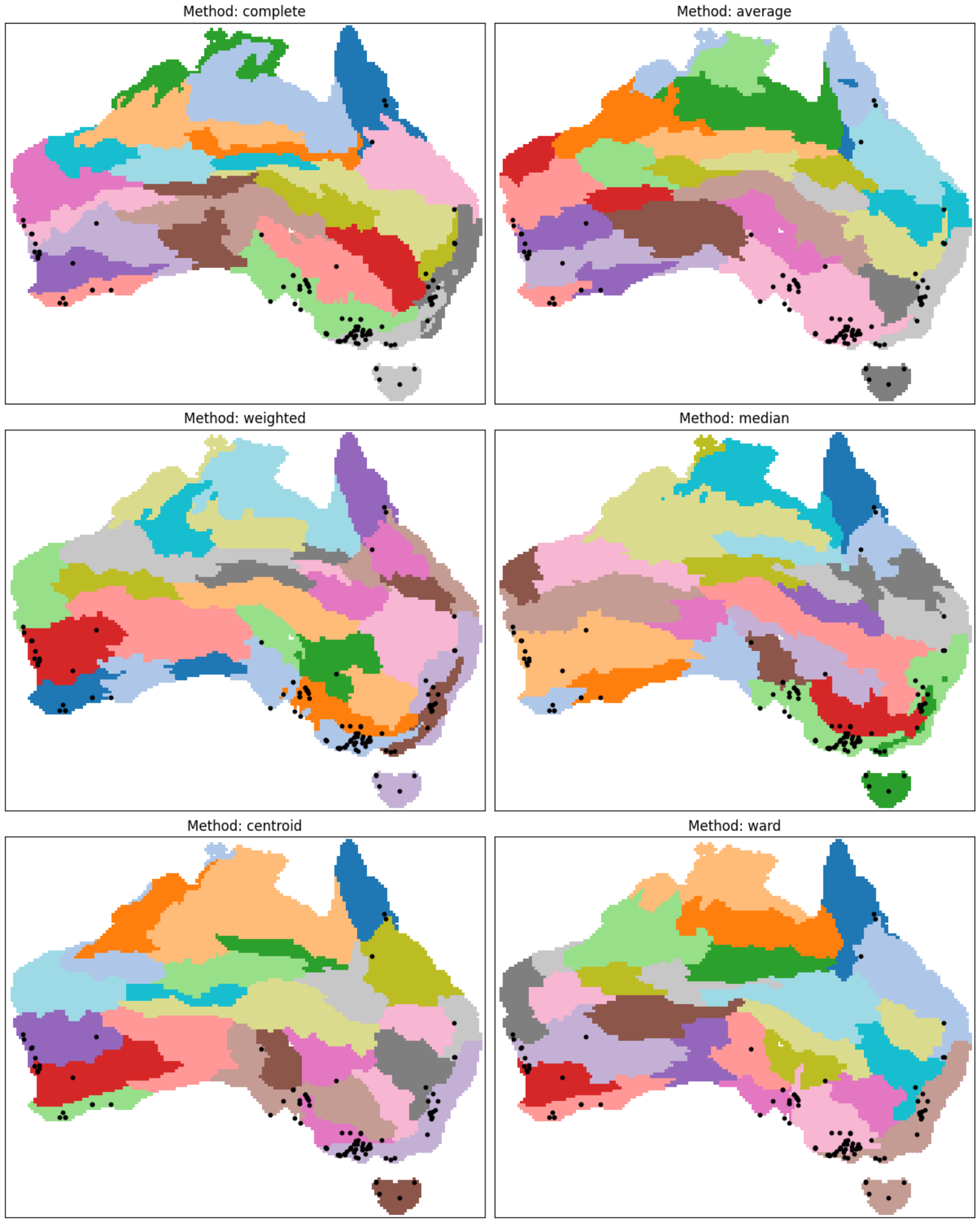
Clusters

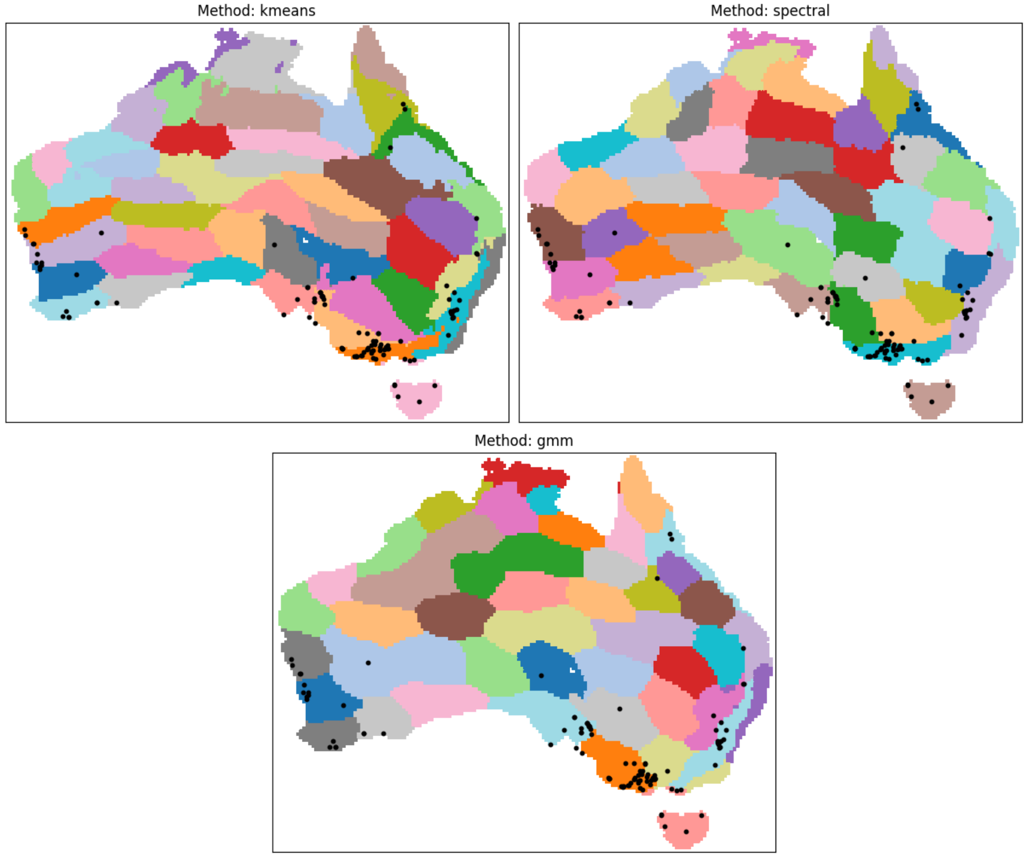




Extra

Clusters





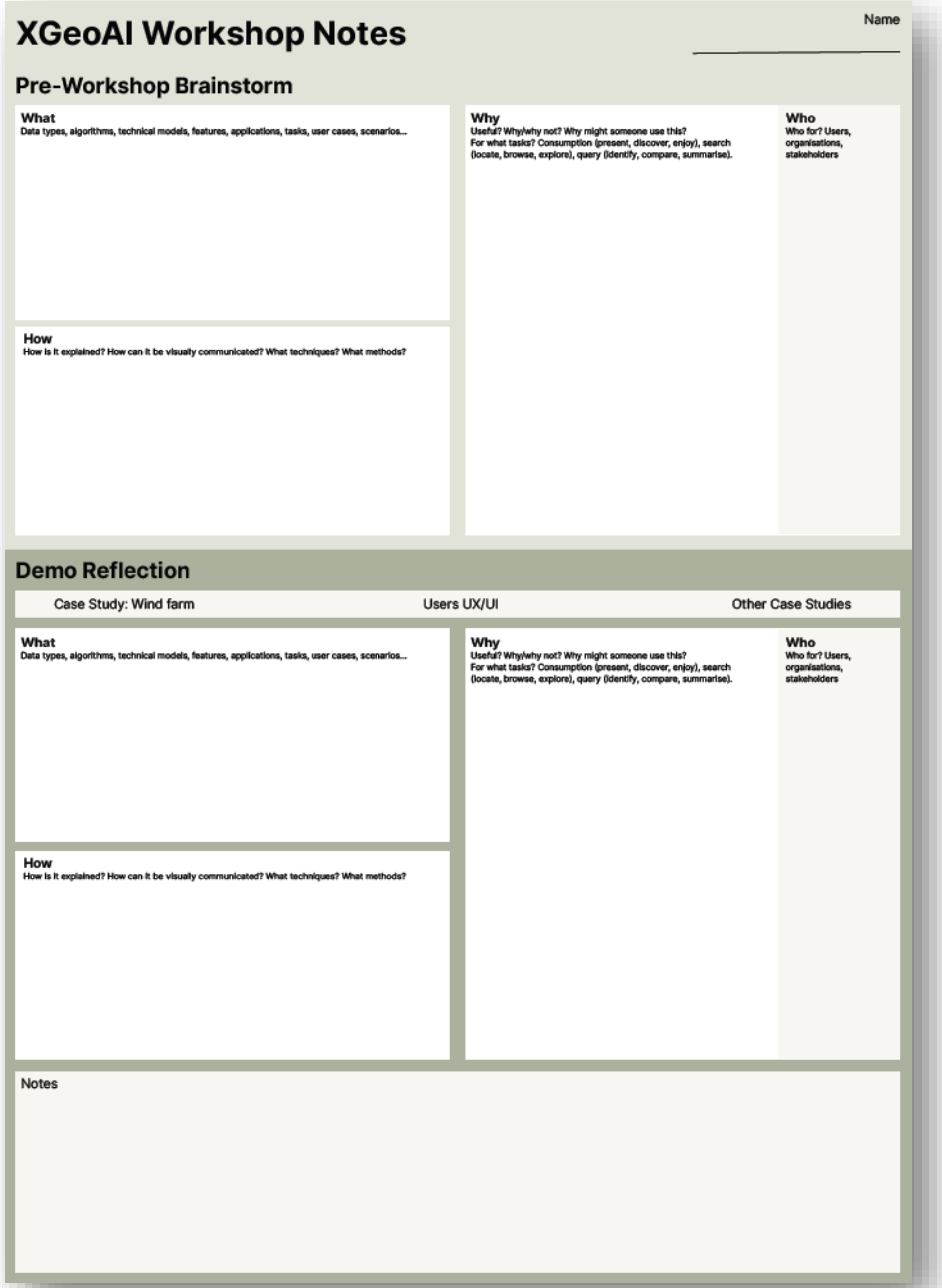
Extra

Clusters

Clustering Method	Top 3 of 15 Clusters (%)	Top 10 of 50 Clusters (%)
Hierarchical - Complete	81.6	91.3
Hierarchical - Median	81.6	87.4
Hierarchical - Average	75.7	93.2
Hierarchical - Ward	81.6	88.3
Hierarchical - Centroid	85.4	87.4
Hierarchical - Weighted	77.7	93.2
K-means	75.7	86.4
Spectral Clustering	71.8	87.4
Gaussian Mixture Model	74.8	93.2

Extra

Workshop



XGeoAI Workshop Notes

Name

Pre-Workshop Brainstorm

What
Data types, algorithms, technical models, features, applications, tasks, user cases, scenarios...

How
How is it explained? How can it be visually communicated? What techniques? What methods?

Why
Useful? Why/why not? Why might someone use this?
For what tasks? Consumption (present, discover, enjoy), search (locate, browse, explore), query (identify, compare, summarise).

Who
Who for? Users, organisations, stakeholders

Demo Reflection

Case Study: Wind farm | Users UX/UI | Other Case Studies

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Data types, algorithms, technical models, features, applications, tasks, user cases, scenarios...

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